21ECO103T
Modern Wireless Communication System

COs	COURSE OUTCOMES
CO-1	Discuss the fundamentals of transmission in wireless systems
CO-2	Provide an overview of various approaches to communication networks
CO-3	Study the numerous different generation technologies with their individual pros and cons
CO-4	Discuss about the principles of operation of the different access technologies like FDMA, TDMA, SDMA and CDMA and their pros and cons
CO-5	Learn about the various mobile data services and short-range networks and gain knowledge on Fundamentals

Unit 2- Network concepts

Communication Networks, LANs, MANs, WANs, Circuit switching, Packet switching, ATM Cellular Networks Introduction, Cells, Duplexing, Multiplexing, Voice coding, Multiple Access Techniques: FDMA, TDMA, SDMA, CDMA, Spectral efficiency

Text books, references:

1. Simon Haykin, David Koilpillai, Michael Moher," Modern Wireless Communication", 1/e, Pearson Education, 2011.
2. Rappaport T.S, "Wireless Communications: Principles and Practice", 2nd edition, Pearson education.

CO-2	Provide an overview of various approaches to communication networks

LAN Basics, MANs, WANs

❖Overview

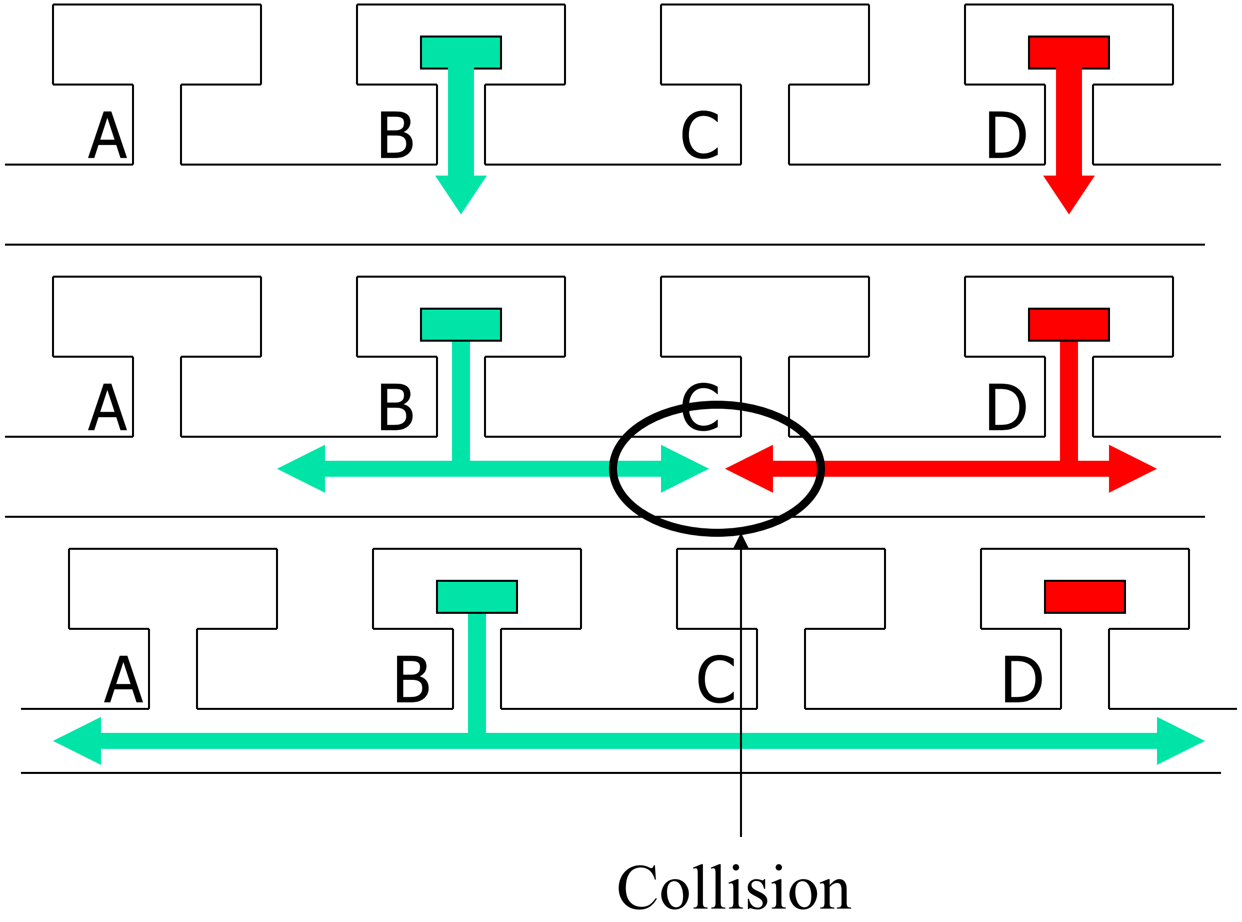
Computer Networks (continued)

- Carrier Sense Multiple Access with Collision Detection (CSMA/CD).
- Types of LANs
- MANs
- WANs
- Network Interconnection Components
- The OSI Model

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

- Usually used in a bus topology
- Used in *Ethernet* LAN’s
- Unlike the token ring, all nodes can send whenever they have data to transmit
- When a node wants to transmit information, it first “listens” to the network. If no one is transmitting over the network, the node begins transmission
- It is however possible for two nodes to transmit simultaneously thinking that the network is clear
- When two nodes transmit at the same time, a *collision* occurs
- The first station to detect the collision sends a jam signal into the network
- Both nodes back off, wait for a random period of time and then re-transmit

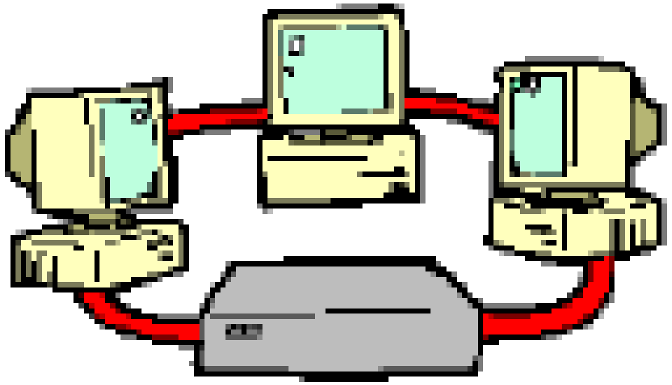
CSMA/CD



Types of LANs

- The three most popular types of LANs are:
  - Token ring network
  - FDDI (Fiber Distributed Data Interface) network
  - Ethernet

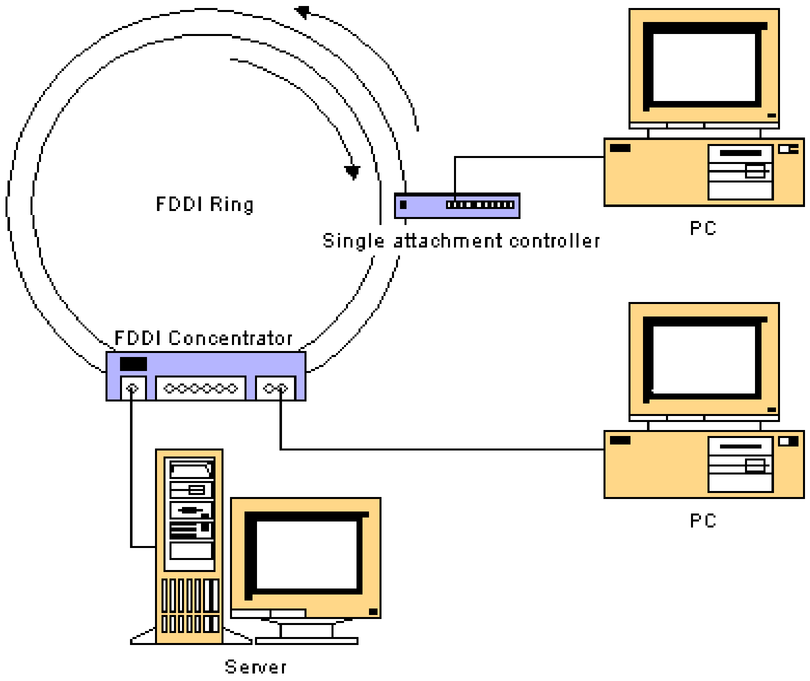
Token Ring Network



- Originally developed by **IBM in 1970's**
- Still IBM's primary LAN technology. **IEEE 802.5**.
- In cases of heavy traffic, the token ring network has higher throughput than ethernet due to the deterministic (non-random) nature of the medium access
- Is used in applications in which delay when sending data must be predictable
- Is a robust network i.e. it is fault tolerant through fault management mechanisms
- Can support data rates of around 16 Mbps
- Typically uses twisted pair

FDDI (Fiber Distributed Data Interface)

- *FDDI* is a standard developed by the American National Standards Institute (ANSI) for transmitting data on optical fibers
- Supports transmission rates of up to 200 Mbps
- Uses a dual ring
  - First ring used to carry data at 100 Mbps
  - Second ring used for primary backup in case first ring fails
  - If no backup is needed, second ring can also carry data, increasing the data rate up to 200 Mbps
- Supports up to 1000 nodes
- Has a range of up to 200 km



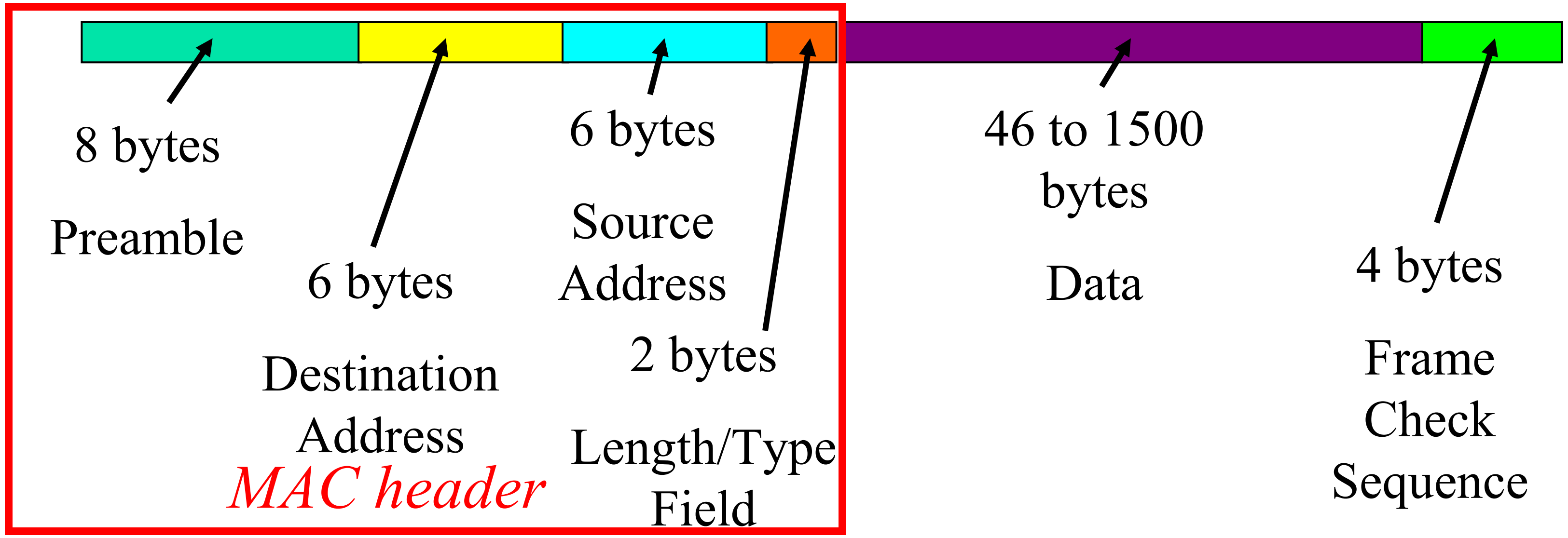
Source:http://burks.brighton.ac.uk/burks/pcinfo/hardware/ethernet/fddi.htm

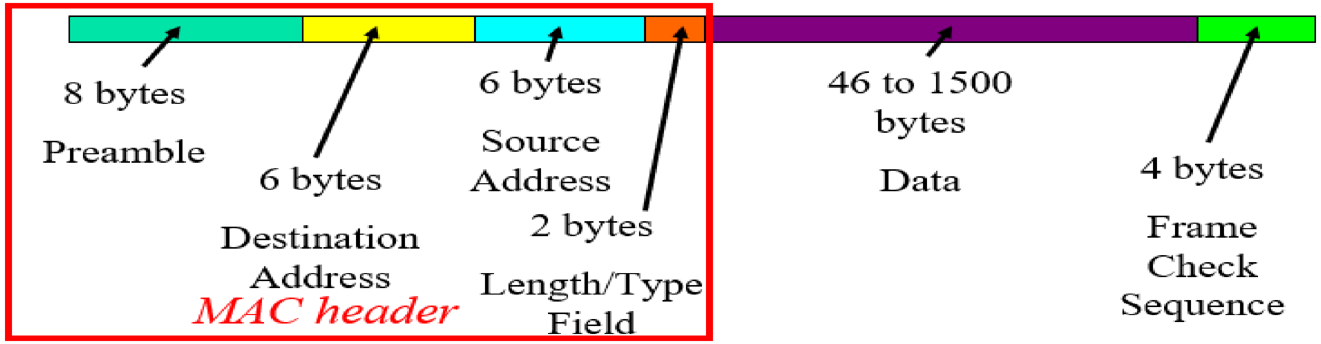
Ethernet

- First network to provide CSMA/CD
- Developed in 1976 by Xerox PARC (Palo Alto Research Center) in cooperation with DEC and Intel
- Is a fast and reliable network solution
- One of the most widely implemented LAN standards
- Can support data rates in the range of 10Mbps- 10 Gbps
- Used with a bus or star topology

The Ethernet Datagram (frame)

- Ethernet traffic is transported in units of a frame called the *Ethernet Datagram*
- The frame consists of a set of bits organized into several fields





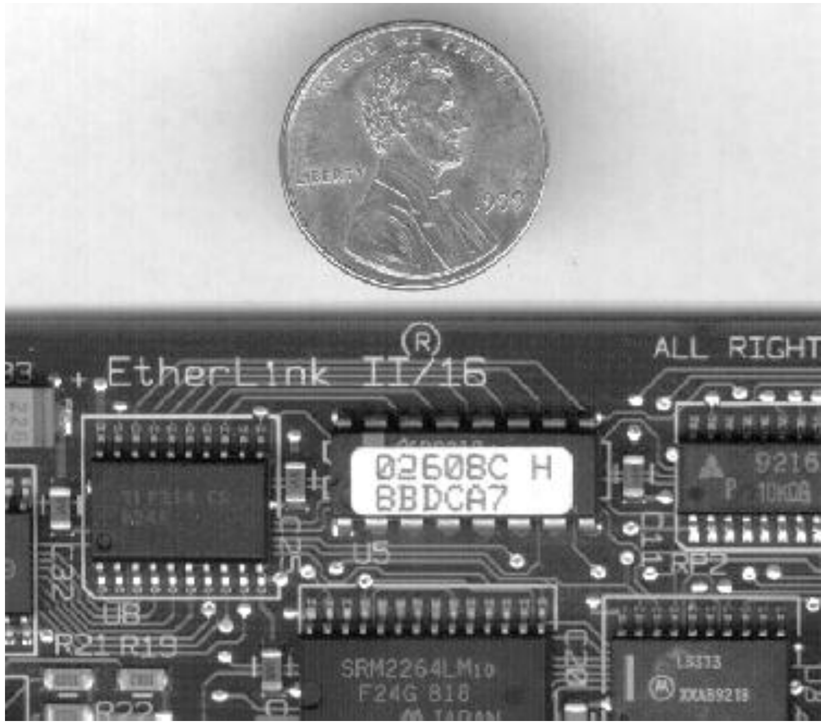
- *Preamble:* Repeating Flag that ID's the sequence as an Ethernet datagram (10101010 7 times followed by 10101011) which is used in synchronizing and alerting the NIC
- *Destination Address:* Unique identifier found on the Network Interface Card that identifies the **recipient**
- *Source Address:* Unique identifier found on the Network Interface Card that identifies the **sender**
- *Length/Type Field:* Tells the recipient what kind of datagram is being received (IP, UDP, etc) and the length of the data
- *Data:* What sort of data is being sent:46 to 1500 bytes (text, JPEG, MP3, etc)
- *Frame Check Sequence:* Error detecting codes (If an error is detected, the frame is discarded)

Padding & Overhead

- The minimum length of data in an ethernet frame should be 46 bytes
- If the length of the message that you want to send is less than 46 bytes, then“padding” is added
- These are extra bits added to bring the total of the message length up to 46 bytes
- The bytes in a frame that do not constitute the actual message that we are interested in sending are called *overhead*
  - The Ethernet frame has 26 bytes of overhead (8+6+6+2+4)
  - If you had 100 bytes of data to send, you’d have to send 126 bytes of data
  - How much overhead is transmitted within the 126 bytes of data?
    - 26/126 = 21%

Ethernet NIC

- The Network Interface card allows computers to communicate with each other through the network
- Each NIC has a 48-bit unique hexadecimal address called the *MAC address*
- A computer or device on a network can be reached by its MAC address through the NIC card
- An example of a MAC address: A1B2C3D4E5F6
- The first 6 hex digits in the MAC address is the OUI (organizationally unique identifier), assigned by the IEEE to each manufacturer (e.g. Cisco, Intel etc). The rest of the MAC address can be assigned in any way by the manufacturer to the individual networking devices that it manufactures.



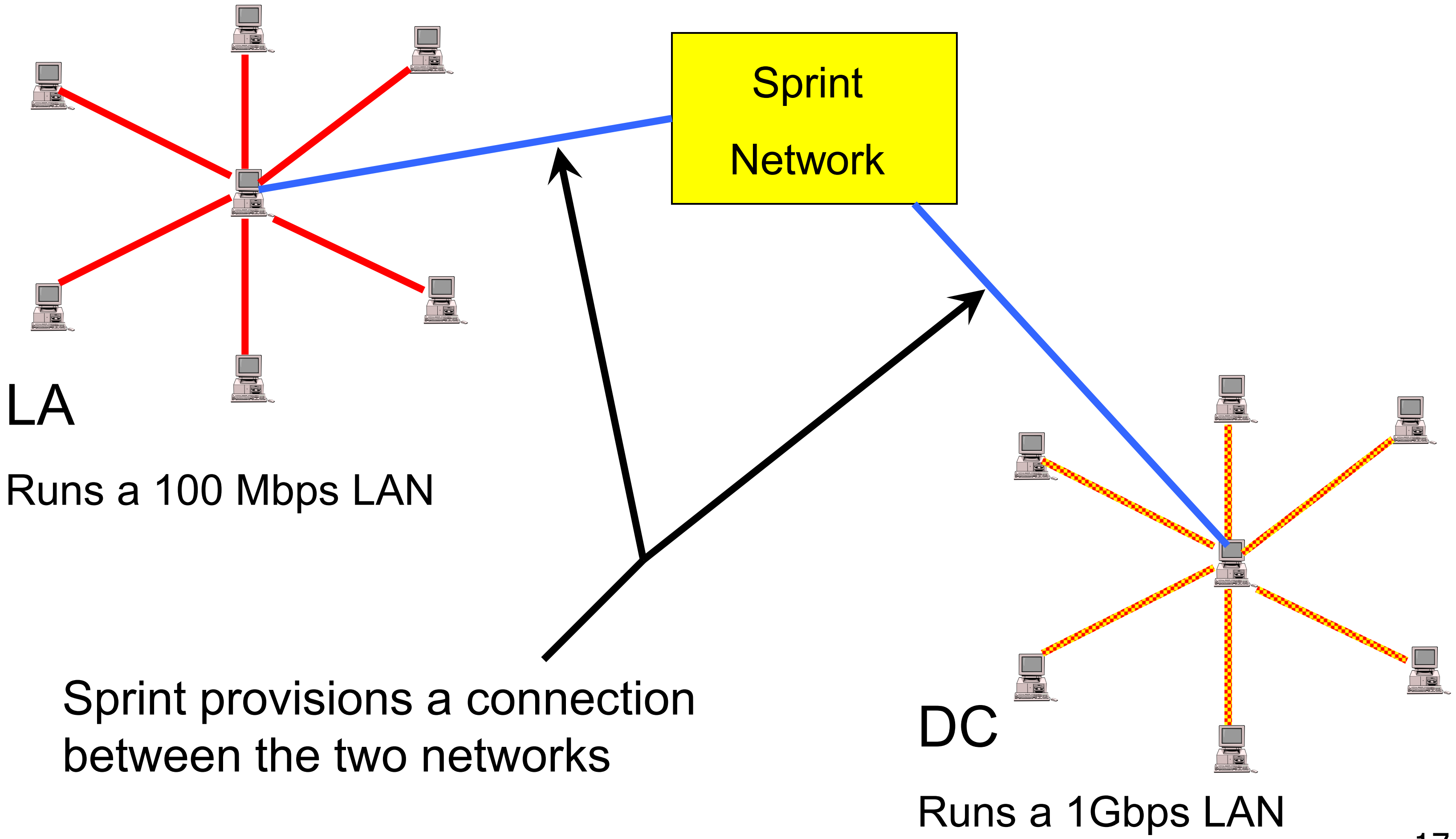
Metropolitan area network

- A *Metropolitan Area Network* (MAN) is a network that is utilized across multiple buildings
- Commonly used in school campuses or large companies with multiple buildings
- Is larger than a LAN, but smaller than a WAN
- Is also used to mean the interconnection of several LANs by bridging them together. This sort of network is also referred to as a *campus network*

Wide area network

- *A Wide Area Network* is a network spanning a large geographical area of around several hundred miles to across the globe
- May be privately owned or leased
- Also called "enterprise networks" if they are privately owned by a large company
- It can be leased through one or several carriers (ISPs-Internet Service Providers) such as AT&T, Sprint, Cable and Wireless
- Can be connected through cable, fiber or satellite
- Is typically slower and less reliable than a LAN
- Services include internet, frame relay, ATM (Asynchronous Transfer Mode)

Example of WAN application



Network Interconnection Components

- Networks can be connected to each other through several components
  - Repeater
  - Bridge
  - Router
  - Gateway
- Before explaining the above components, we need to understand the OSI model

The OSI Model

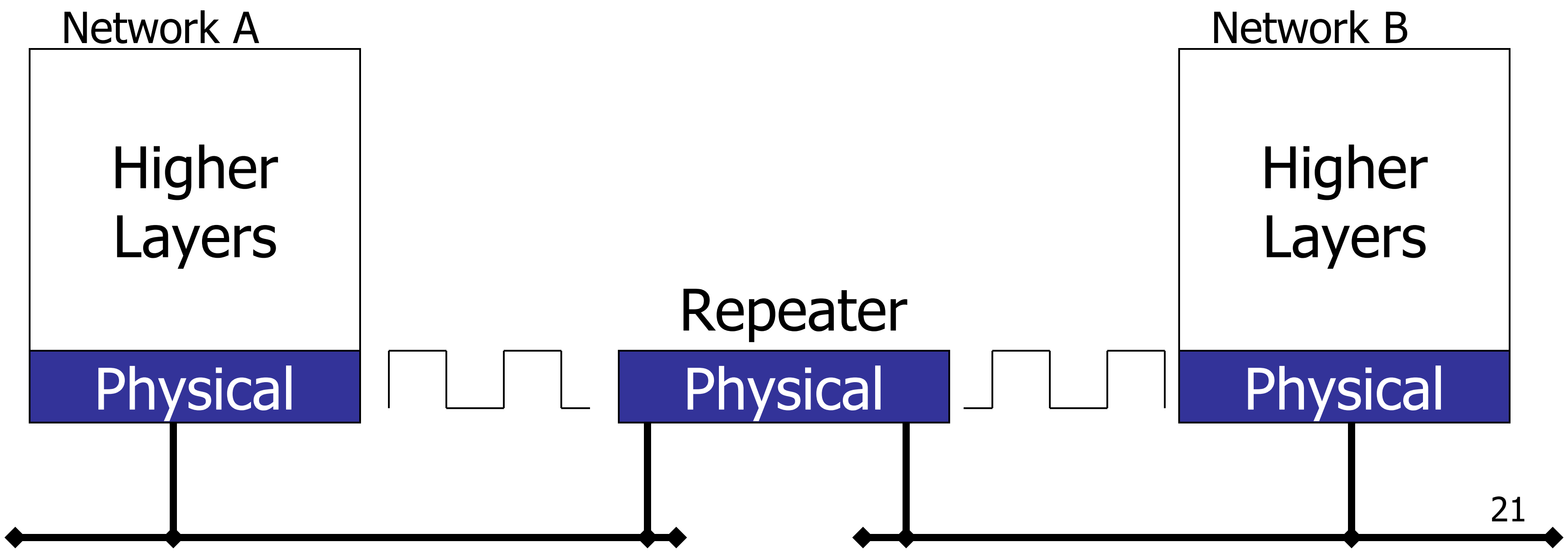
- The Open Systems Interconnection (OSI) model is a theoretical framework for understanding and explaining networking protocols
- Originally an effort by the ISO (International Standards Organization) to standardize network protocols
- TCP/IP became the dominant set of standards but the OSI model is widely used to help understand protocols
- The OSI model defines 7 layers of functional communications protocols.

The OSI Model

7	Application Layer	*Provides a network interface for applications*
6	Presentation Layer	*Translates data to standard format*
5	Session Layer	*Establishes sessions between computers*
4	Transport Layer	*Provides error control and flow control*
3	Network Layer	*Supports logical addressing and routing*
2	Data Link Layer	*Interfaces with network adapter*
1	Physical Layer	*Converts information into transmitted pulses*

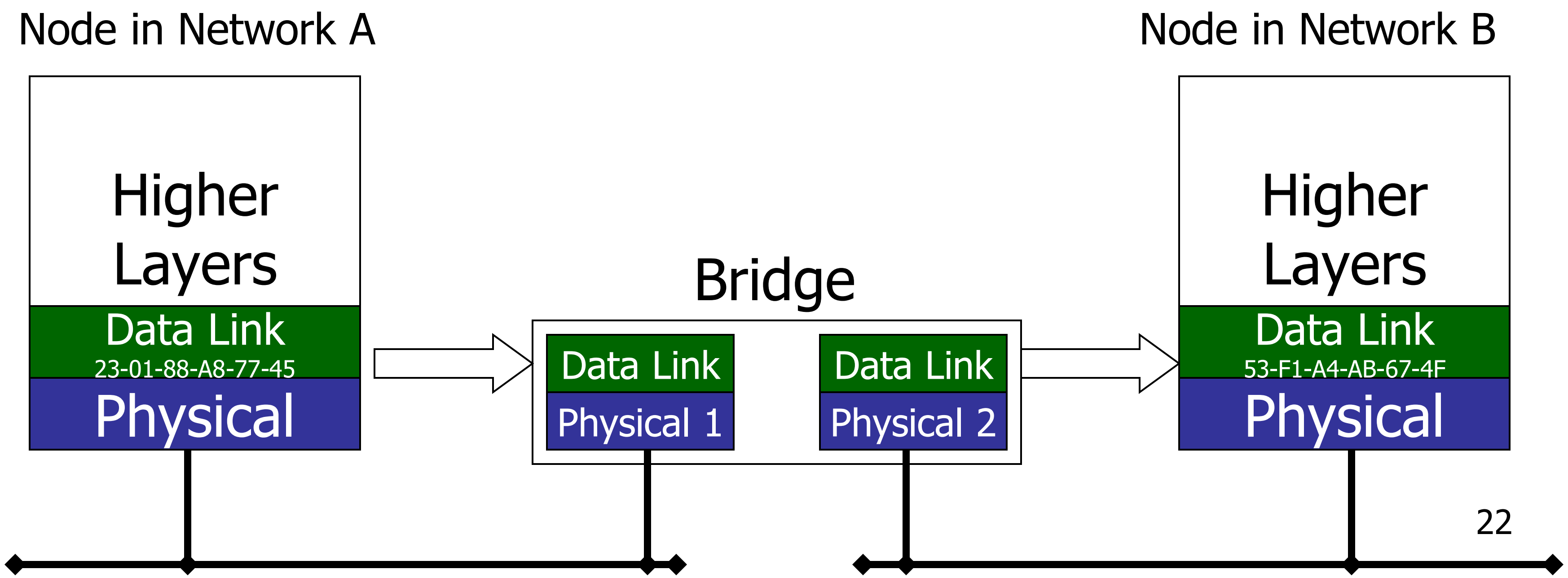
Repeater

- Regenerates and propagates all electrical transmissions between 2 or more LAN segments
- Allows extension of a network beyond physical length limitations
- Layer 1 of the “OSI model”



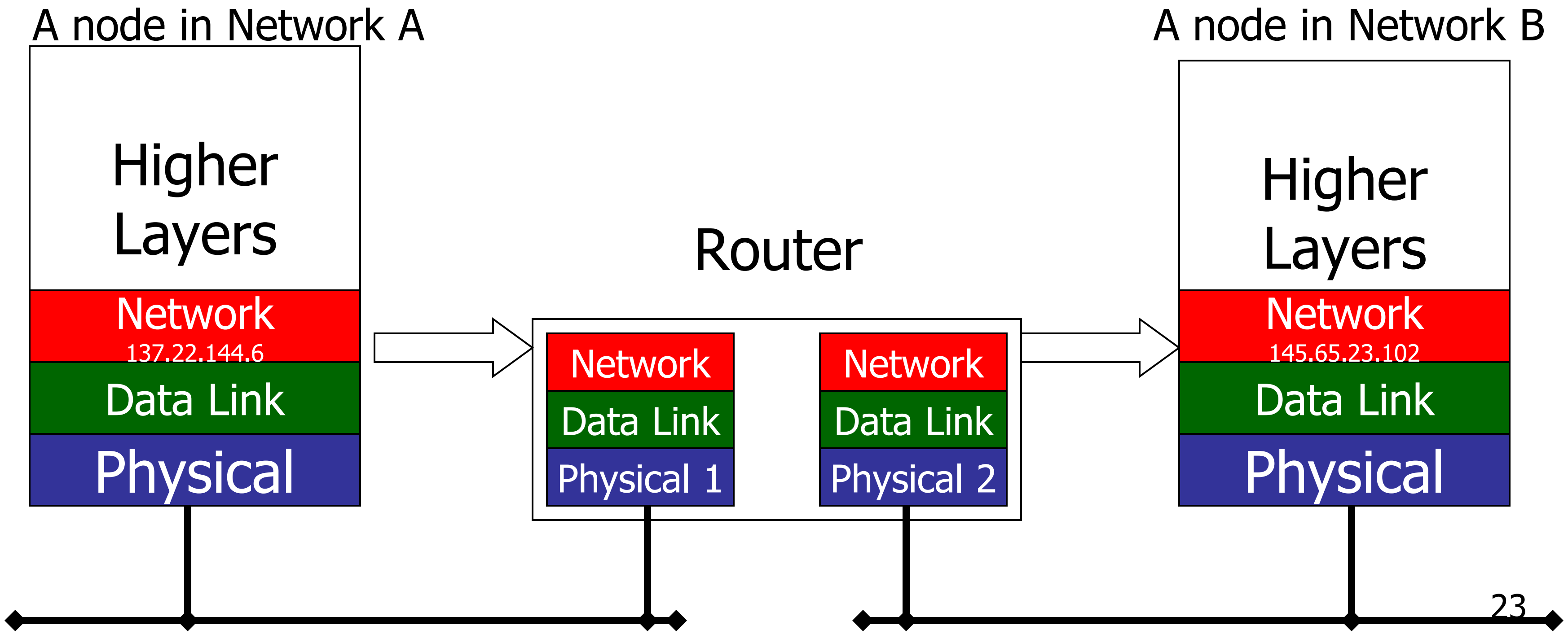
Bridge

- Connects 2 or more LAN segments and uses data link layer addresses (e.g.MAC addresses) to make data forwarding decisions
- Copies frames from one network to the other
- Layer 2 of the "OSI model"



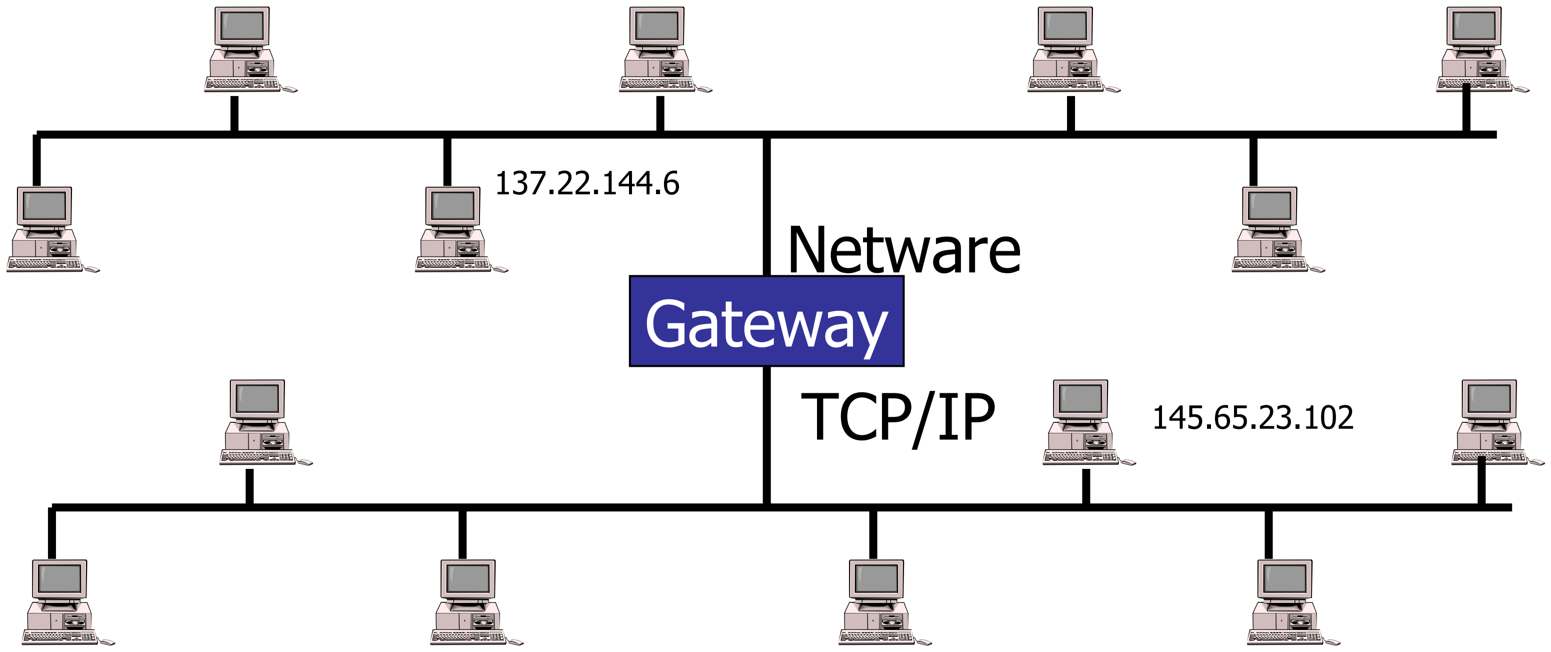
Router

- Connects 2 or more networks and uses network layer addresses (like IP address) to make data forwarding decisions
- Layer 3 of the "OSI model"



Gateway

- Connects 2 or more networks that can be of different types and provides protocol conversion so that end devices with dissimilar protocol architectures can interoperate



Introduction to Data Communication and Networking

Objectives :

✓ To Introduce the students to Switching Types, Network Categories and Network Topology

- The main objective of data communication and networking is to enable seamless exchange of data between any two points in the world.
- This exchange of data takes place over a computer network.
- To understand the concepts of various switching techniques, types of networks and topology.

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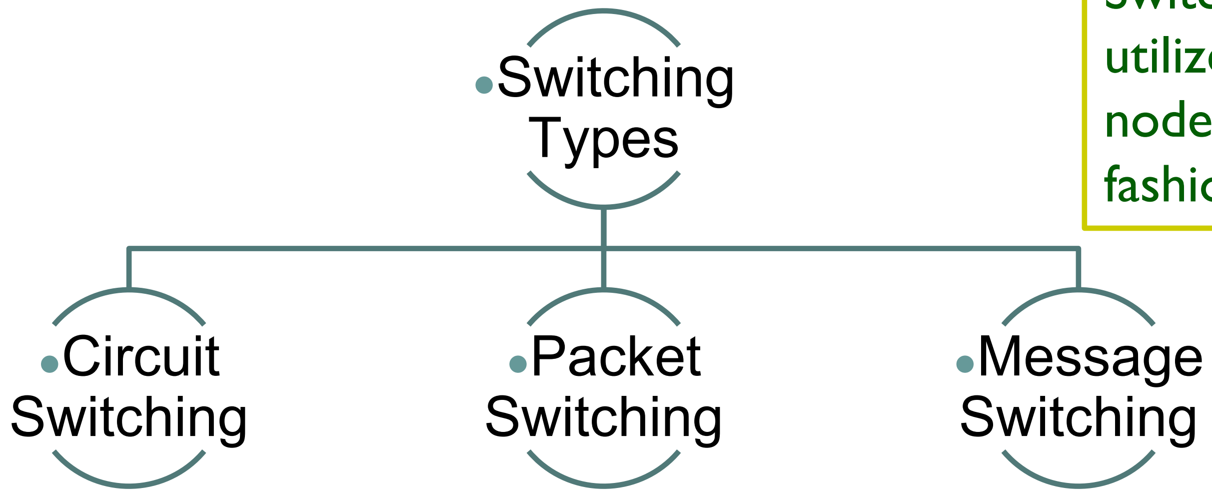
Switching Types

Switching is the generic method for establishing a path for point-to-point communication in a network.

- One of the most important functions of the network layer is to employ the switching capability of the nodes in order to route messages across the network.
- In a switched network, some of these nodes are connected to the end systems and others for routing.



Switching involves the nodes in the network to utilize their direct communication lines to other nodes so that a path is established in a piecewise fashion.

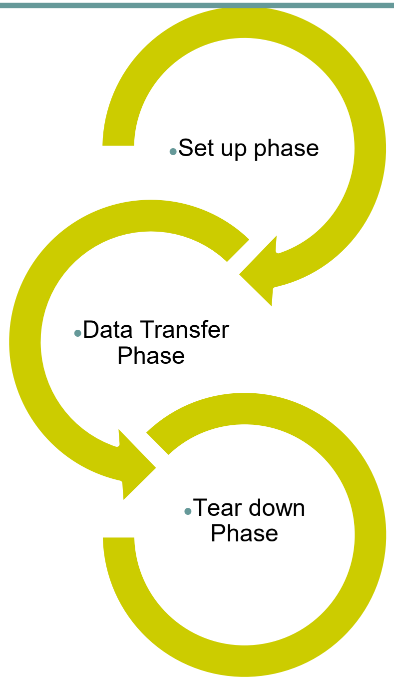
Circuit Switching

In circuit switching, two communicating stations are connected by a dedicated communication path, which consists of intermediate nodes in the network and the links that connect these nodes.

A circuit-switched network consists of a set of switches connected by one or more dedicated physical links.

The end systems such as computers or telephones are directly connected to a switch.

Three phases to connect and transfer the information



Each link is normally divided into n channels by using FDM or TDM.

Circuit Switching

Connection Establishment / Setup :

Connection is established by a dedicated circuit (combination of channels in links) for communication between devices

Data transfer

It is from the source to the destination. of the network.

The data may be analog or digital, depending on the nature

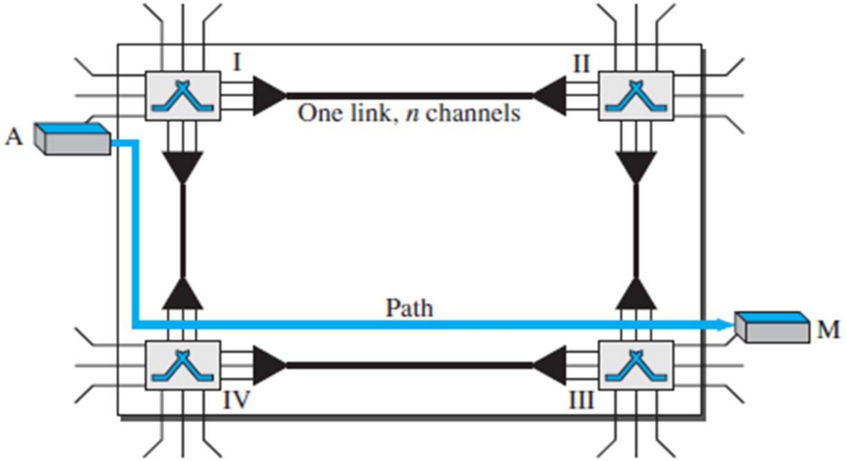
The connection is generally full-duplex

Connection Release / Tear down

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Phases in Circuit Switching

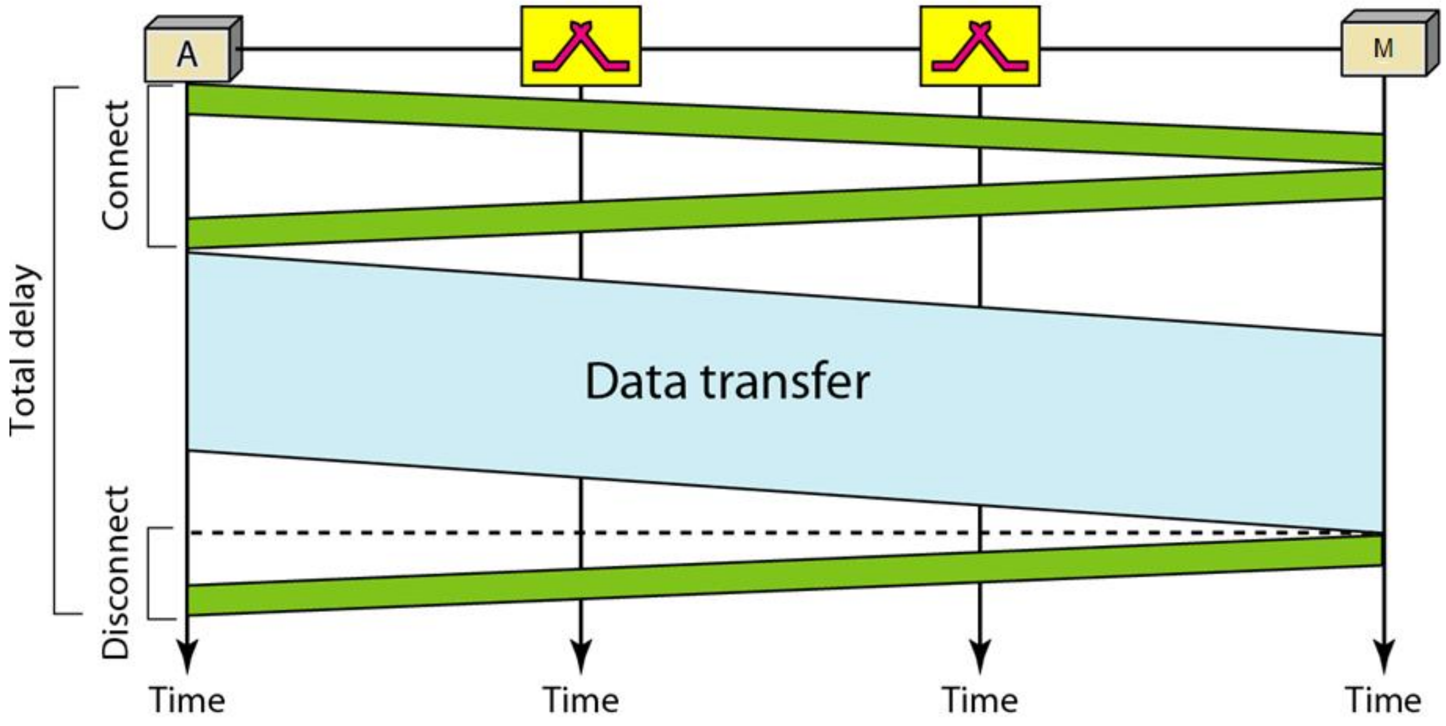
Circuit switching - technique used in telephony, where the caller sends a special message with the address of the callee (i.e. by dialing a number) to state its destination.

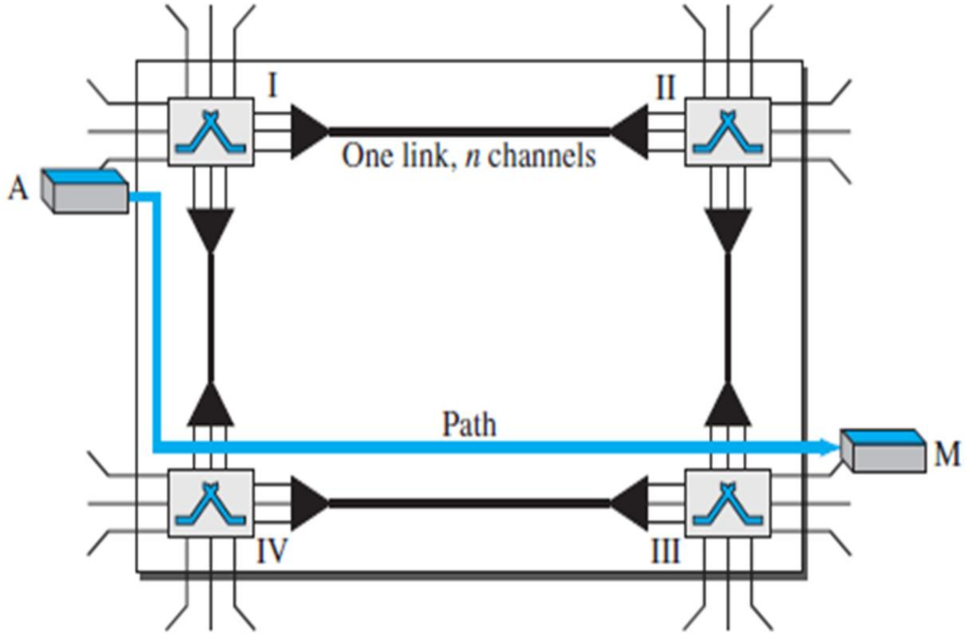


Circuit switching between A and M via Switch I – IV - III

An end-to-end addressing is required for creating a connection between the two end systems. It could be addresses of the computers in a TDM network, or telephone numbers in an FDM network

Circuit Switching Technique





Circuit switching between A and M via Switch I – IV - III

when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I, which find the route to destination.

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Circuit Switching Technique

Efficiency :

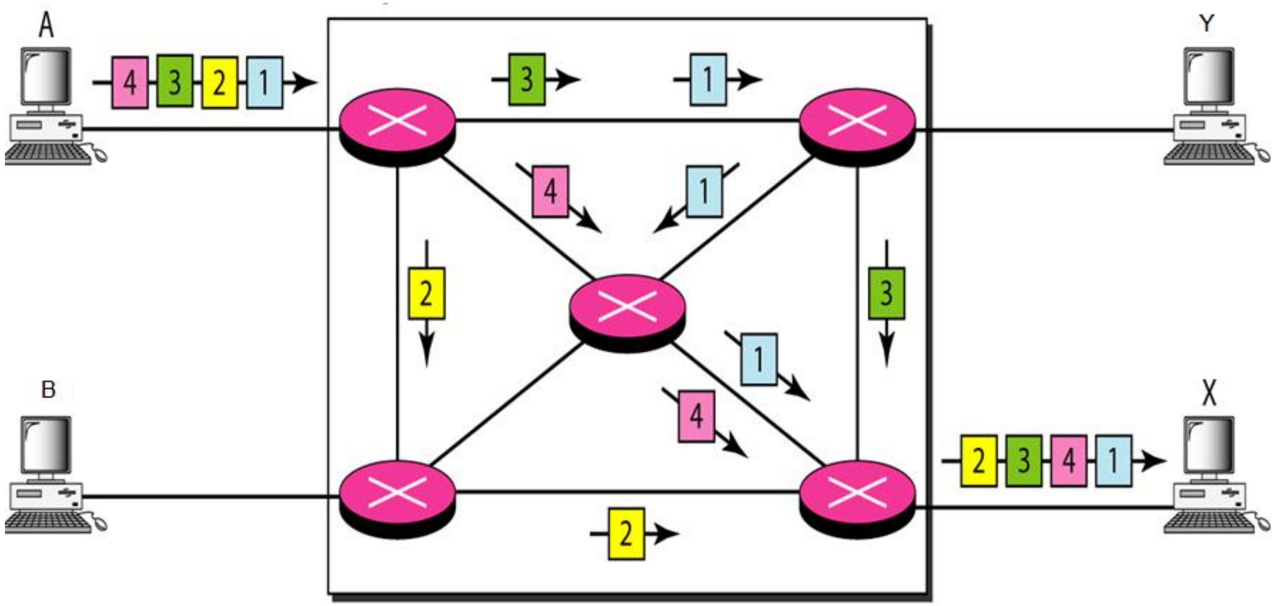
- *The resources are allocated during the entire duration of the connection, so not efficient.*
- *These resources are unavailable to other connections.*
- *In a telephone network, people normally terminate the communication when they have finished their conversation.*
- *However, in computer networks, a computer can be connected to another computer even if there is no activity for a long time*

Delay:

- *Minimal delay (coz of dedicated resources).*
- *The total delay is due to the time needed to create the connection, transfer data, and disconnection.*
    - *The **delay caused by the setup** is the sum of propagation time of the source computer request, the request signal transfer time, the propagation time of the acknowledgment from the destination computer, and the signal transfer time of the acknowledgment.*
    - *The **delay due to data transfer** is the sum of : the propagation time and data transfer time.*
    - *The **delay due to tear down** is maximum, If the receiver requests disconnection.*

Packet Switching

A packet is handed over from node to node across the network. Each receiving node temporarily stores the packet, until the next node is ready to receive it, and then passes it onto the next node. This technique is called store-and-forward and overcomes one of the limitations of circuit switching..



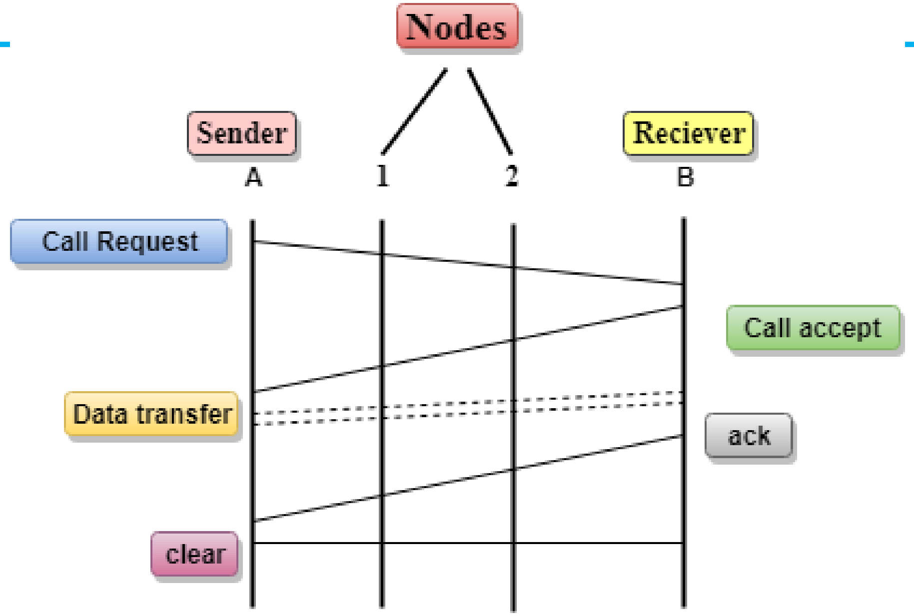
- Packet Switching
  - Virtual Circuit Switching - Connection oriented. Route is established before data via connection request packet to the destination.
  - Datagram Switching – Connectionless. Does not require pre-established route. Each packet is treated independently and it travel along different routes in the network to reach destination.

In a packet-switched network, there is no resource reservation; resources are allocated on demand.

Packet Switching Technique

Virtual Circuit Switching

- In Virtual circuit switching, a preplanned route is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.



Datagram Packet switching:

- Packet is known as a datagram, is considered as an independent entity.
- Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- The path is not fixed. Intermediate nodes take the routing decisions to forward the packets.

Packet Switching Technique

Advantages of Packet Switching:

- **Cost-effective:** In packet switching technique, switching devices do not require massive secondary storage to store the packets, so cost is minimized to some extent.
- **Reliable:** If any node is busy, then the packets can be rerouted.
- **Efficient:** As it does not require any established path prior to the transmission, and many users can use the same communication channel simultaneously, hence makes use of available bandwidth very efficiently.

Disadvantages of Packet Switching:

- Cannot implemented in those applications that require low delay and high-quality services
- The protocols used are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets.
- It can also lead to the loss of critical information if errors are nor recovered.

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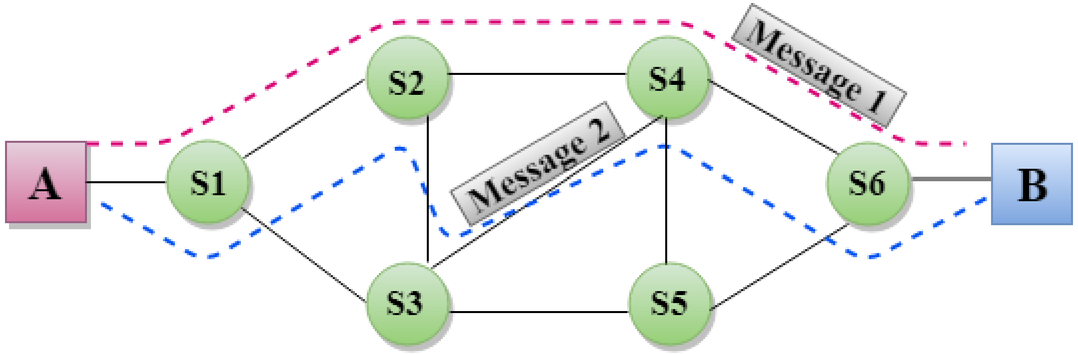
Message Switching

Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.



- There is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as store and forward network.

Message switching treats each message as an independent entity.

Message Switching

Advantages of Message Switching:

- Efficient utilization of resources: Data channels are shared among the communicating devices.
- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages of Message Switching:

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.

Review Questions

Q1. Mark “YES” or :NO”

Comparison between Switching Techniques	Circuit Switching	Message Switching	Packet Switching
Path established in advance	YES / NO		
Store and Forward Technique	YES / NO	YES / NO	YES / NO
Message Follows Multiple Routes	YES / NO		

Comparison between Packet Switching Techniques	Virtual Circuit Switching	Datagram Switching
Nodes takes the routing decision	YES / NO	
Congestion Occurs	YES / NO	

Answers

Q1. Mark “YES” or :NO”

Comparison between Switching Techniques	Circuit Switching	Message Switching	Packet Switching
Path established in advance	YES		
Store and Forward Technique	NO	YES	YES
Message Follows Multiple Routes	NO		

Comparison between Paccket Switching Techniques	Virtual Circuit Switching	Datagram Switching	
Nodes takes the routing decision	NO		
Congestion Occurs	YES		

Categories of Network - LAN, MAN & WAN

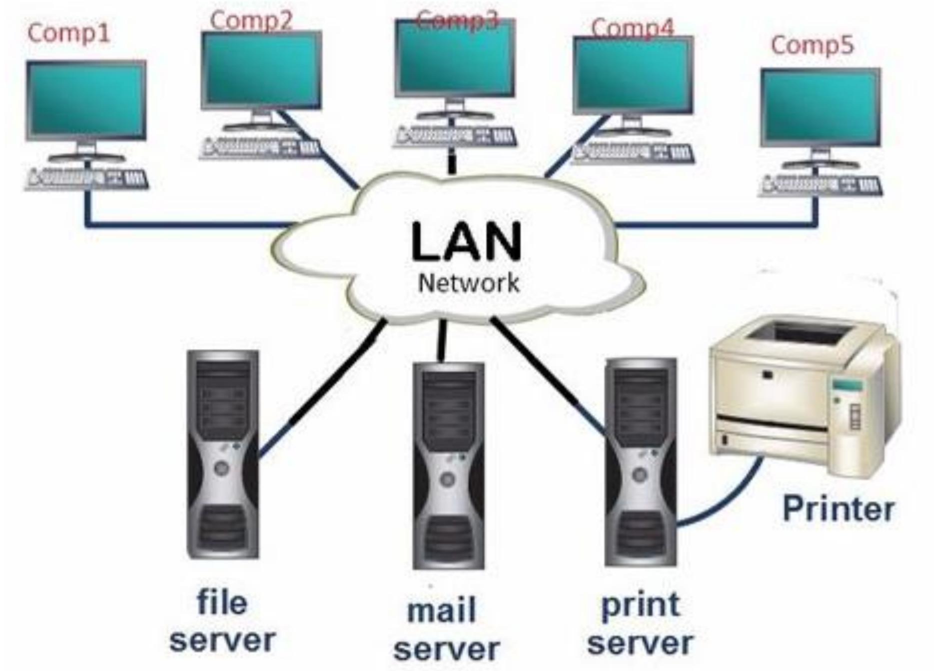
Networks are categorized on the basis of their size.

The three basic categories of computer networks are: LAN, MAN & WAN

- Local Area Network (LAN)
  - Limited to a few kilometers of area. Privately owned (Ex. Office building)
- Metropolitan Area Network (MAN)
  - Size between LAN & WAN. It is larger than LAN but smaller than WAN. (Ex. Network connecting City)
- Wide Area Network (WAN)
  - Made of all the networks in a (geographically) large area. (Ex. Network connecting the entire State)

Local Area Network - LAN

- Local Area Network (LAN) is a computer network spanned inside a building and operated under single administrative system.
- LAN covers an organization offices, schools, colleges or universities.
- Number of systems connected in LAN may vary from as least as two to as much as 16 million.
- LANs are composed of inexpensive networking and routing equipment.



Features of LAN

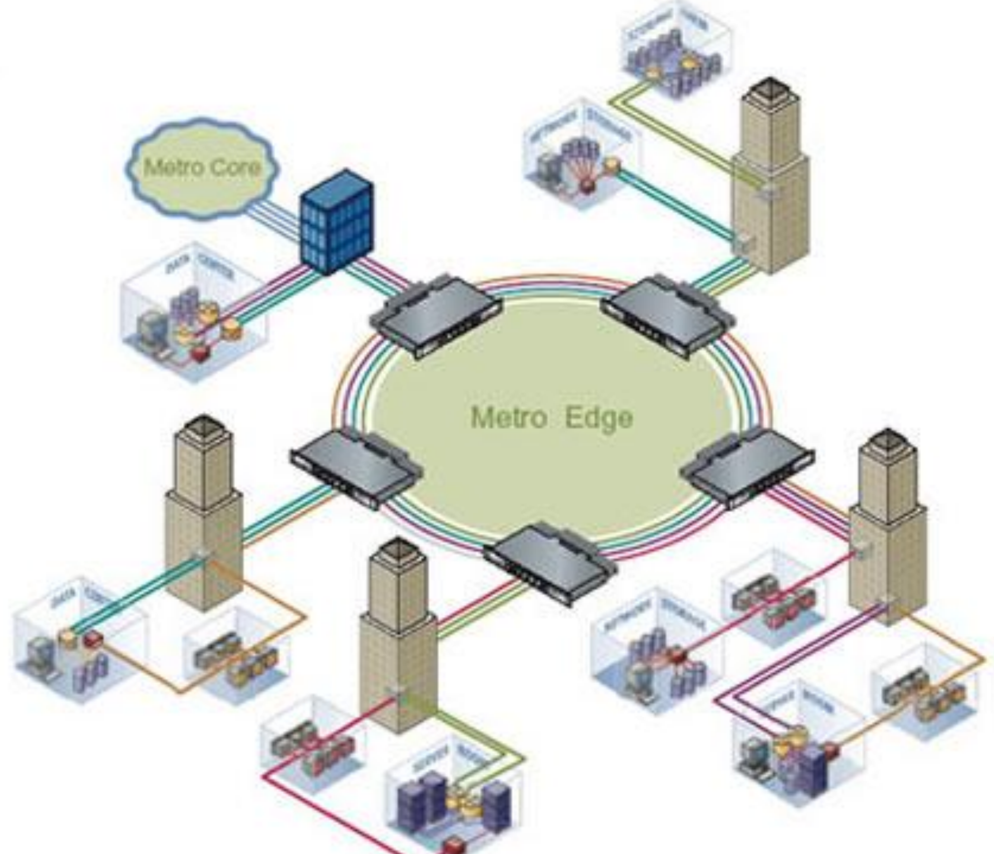
- LAN provides a useful way of sharing the resources between end users
- LAN uses either Ethernet or Token-ring technology. Ethernet is most widely employed and uses Star topology, while Token-ring is rarely used.
- LAN can be wired, wireless, or in both forms at once

LAN Services:

LAN can helps to connect all of its offices in a building premises.

Metropolitan Area Network - MAN

- Metropolitan Area Network (MAN) generally expands throughout a city such as cable TV network.
- It can be in the form of Ethernet, Token-ring, ATM, or Fiber Distributed Data Interface (FDDI).
- Metro Ethernet is a service which is provided by ISPs. This service enables its users to expand their LANs.



Features of MAN

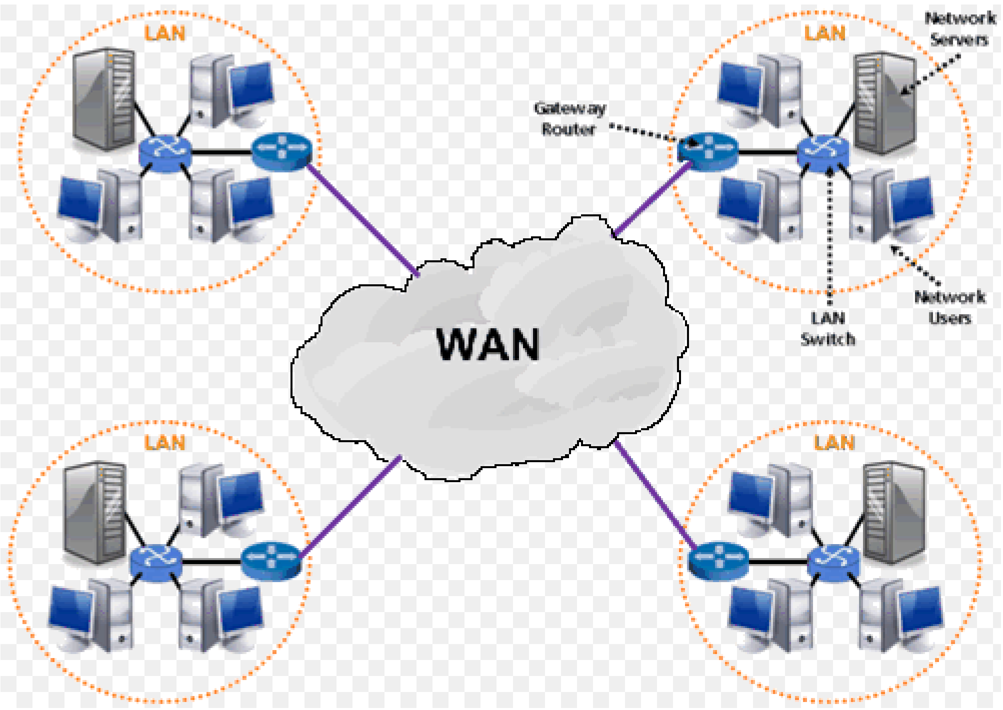
- Backbone of MAN is high-capacity and high-speed fiber optics.
- MAN works in between Local Area Network and Wide Area Network.
- MAN provides uplink for LANs to WANs or internet

MAN Services:

MAN can help an organization to connect all of its offices in a city.

Wide Area Network - WAN

- Computer network that covers a large geographical area comprising a region, a country, a continent or even the whole world.
- WAN includes the technologies to transmit data, image, audio and video information over long distances and among different LANs and MANs.



Features of WAN

- WANs have a large capacity, connecting a large number of computers over a large area, and are inherently scalable.
- They facilitate the sharing of regional resources.
- They provide uplinks for connecting LANs and MANs to the Internet.
- Communication links are provided by public carriers like telephone networks, network providers, cable systems, satellites etc.
- Typically, they have low data transfer rate and high propagation delay.
- They generally have a higher bit error rate.

WAN Services:
- The Internet
- 4G Mobile Broadband Systems
- A network of bank cash dispensers

Comparison- LAN, MAN & WAN

Key Parameters	LAN	MAN	WAN
Ownership	Owned by private organizations.	Ownership can be private or public.	Ownership can be private or public.
Speed	LAN speed is quiet high.	MAN speed is average.	WAN speed is lower than that of LAN.
Delay	Network Propagation Delay is short.	Network Propagation Delay is moderate.	Network Propagation Delay is longer.
Congestion	LAN has low congestion as compared to WAN.	MAN has higher congestion than LAN.	WAN has higher congestion than both MAN and LAN.
Fault Tolerance	Fault Tolerance of LAN is higher than WAN.	Fault Tolerance of MAN is lower than LAN.	Fault Tolerance of WAN is lower than both LAN and MAN.
Maintenance	Designing and maintaining LAN is easy and less costly than WAN.	Designing and maintaining WAN is complex and more costly than LAN.	Designing and maintaining WAN is complex and more costly than both.

Review Questions

Q1. *Computer that spans a limited physical area, usually ranging from a small office to a building is known as _________________.*

Q2. Multiple LANs can be connected to form a single MAN (Metropolitan Area Network). State TRUE/FALSE.

Q3. To form a WAN or MAN network, public networks can be used in between. State TRUE or FALSE.

Q4. Which cable between Twisted-Pair-Cable (TPC) and Coaxial-Cable (CC) work for transmitting data to more distances?

Q5. Mr. John is a small business man who runs a Hardware, he has been experiencing a problem with his small accounting department which he depends on to provide him with the sales reports.

Mr John wants to share information between his 7 computer station and have on central printing area, what type of network would you recommend to Mr. John?

Answers

- 1. Local Area Network
- 2. True
- 3. True
- 4. Coaxial Cable
- 5. LAN

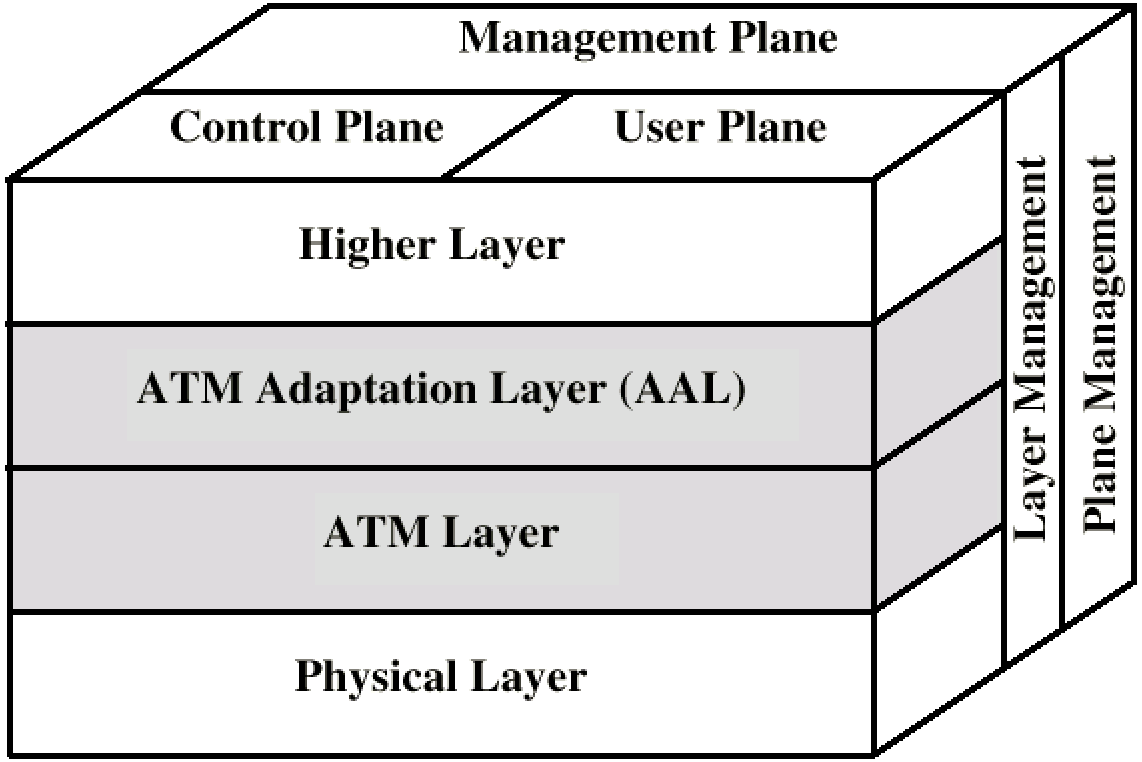
William Stallings
Data and Computer Communications

Chapter 11
Asynchronous Transfer Mode and Frame Relay

Protocol Architecture

- Similarities between ATM and packet switching
  - Transfer of data in discrete chunks
  - Multiple logical connections over single physical interface
- In ATM flow on each logical connection is in fixed sized packets called cells
- Minimal error and flow control
  - Reduced overhead
- Data rates (physical layer) 25.6Mbps to 622.08Mbps

Protocol Architecture (diag)



Introduction to ATM
Slide Content:
•Asynchronous Transfer Mode (ATM) is a **cell-based switching technology**
•Designed for **high-speed, integrated voice, video, and data**
•Uses **fixed-size cells (53 bytes)** for predictable performance
•Widely used in **B-ISDN, backbone networks, and early broadband systems**
Key Point: ATM separates functionality into **layers and planes** for flexibility and QoS.

ATM Protocol Architecture – Overview

ATM architecture is divided into:

- **Three Layers**
  - Physical Layer
  - ATM Layer
  - ATM Adaptation Layer (AAL)
- **Three Planes**
  - User Plane
  - Control Plane
  - Management Plane

•Supports **QoS, traffic management, and scalability**

Physical Layer

- Responsible for **actual transmission of bits**
- Handles:
    - Line coding
    - Bit timing
    - Signal synchronization
- Interfaces ATM cells with physical media

Examples:

- SONET/SDH
- Optical fiber
- Copper links

Function: Ensures reliable **bit-level transport**

ATM Layer

•Core layer of ATM architecture

•Functions:

- Cell multiplexing and demultiplexing
- Header generation and extraction
- Virtual Path Identifier (VPI) & Virtual Channel Identifier (VCI) handling
- Traffic shaping and congestion control

Key Feature: Works with **fixed 53-byte ATM cells**

ATM Adaptation Layer (AAL)

•Interface between **ATM layer and higher layers**

•Converts variable-length data into ATM cells

•Performs:

- Segmentation (at sender)
- Reassembly (at receiver)

Main AAL Types:

AAL1: Constant Bit Rate (voice)

AAL2: Variable Bit Rate real-time

AAL3/4: Data services (obsolete)

AAL5: IP and data traffic (most common)

Higher Layers

- Includes application and transport protocols
- Examples:
    - IP
    - TCP/UDP
    - Voice, video, and multimedia applications
- Independent of ATM cell structure

Role: Provides **end-user services**

User Plane

•Handles **actual user data transfer**

•Includes:

- AAL
- ATM Layer
- Physical Layer

•Supports:

- Flow control
- Error detection (limited)

Example: Voice call or video stream transmission

Control Plane

•Responsible for **connection setup and release**

•Functions:

- Signaling
- Call control
- Routing decisions

Protocols Used:

•Q.2931

•UNI / NNI signaling

Key Point: No user data flows through this plane

Management Plane

- Provides **network management and maintenance**
- Divided into:
    - **Layer Management:** Manages individual layers
    - **Plane Management:** Coordinates across planes

Functions:

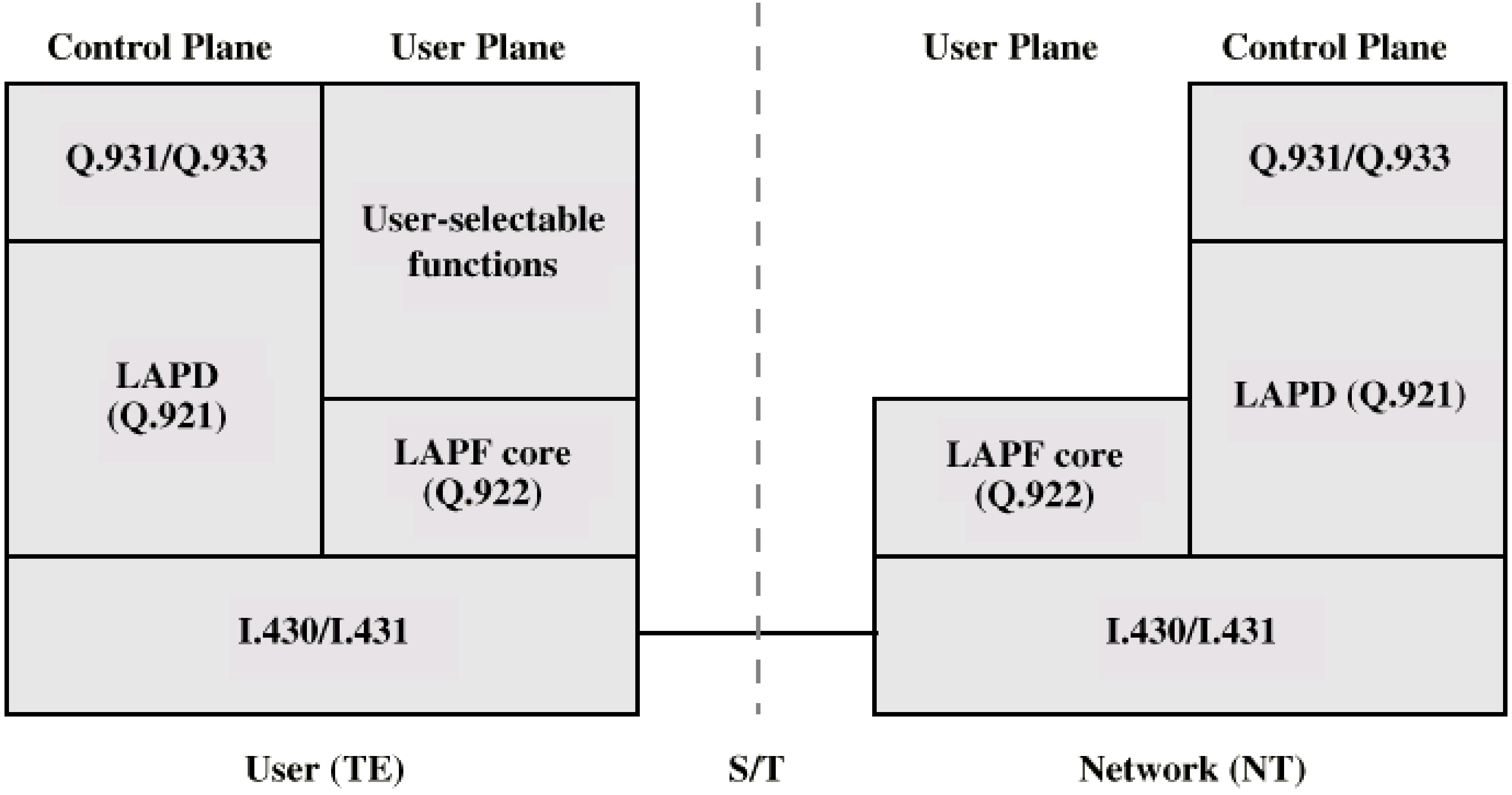
- Fault management
- Performance monitoring
- Configuration management

Features of ATM Architecture

- Fixed-size cells → low latency
- Strong QoS support
- Separation of control and data
- Scalable and flexible design
- Suitable for multimedia traffic

Advantages:
•Predictable performance
•Efficient multiplexing
•QoS guarantees
Limitations:
•High overhead (53-byte cell)
•Complex implementation
•Largely replaced by IP/MPLS

Protocol Architecture



Control Plane

- Between subscriber and network
- Separate logical channel used
  - Similar to common channel signaling for circuit switching services
- Data link layer
  - LAPD (Q.921)
  - Reliable data link control
  - Error and flow control
  - Between user (TE) and network (NT)
  - Used for exchange of Q.933 control signal messages

User Plane

- End to end functionality
- Transfer of info between ends
- LAPF (Link Access Procedure for Frame Mode Bearer Services) Q.922
  - Frame delimiting, alignment and transparency
  - Frame mux and demux using addressing field
  - Ensure frame is integral number of octets (zero bit insertion/extraction)
  - Ensure frame is neither too long nor short
  - Detection of transmission errors
  - Congestion control functions

LAPF Core Formats

Flag	Address	Information	FCS	Flag
<---1---> octet	<-------2-4------>	<-------------Variable-------------->	<--------2------->	<---1-->

(a) Frame format

8	7	6	5	4	3	2	1
Upper DLCI						C/R	EA 0
Lower DLCI				FECN	BECN	DE	EA 1

(b) Address field - 2 octets (default)

8	7	6	5	4	3	2	1
Upper DLCI						C/R	EA 0
DLCI				FECN	BECN	DE	EA 0
Lower DLCI or DL-CORE control						D/C	EA 1

(c) Address field - 3 octets

8	7	6	5	4	3	2	1
Upper DLCI						C/R	EA 0
DLCI				FECN	BECN	DE	EA 0
DLCI							EA 0
Lower DLCI or DL-CORE control						D/C	EA 1

(d) Address field - 4 octets

EA	Address field extension bit
C/R	Command/response bit
FECN	Forward explicit congestion notification
BECN	Backward explicit congestion notification
DLCI	Data link connection identifier
D/C	DLCI or DL-CORE control indicator
DE	Discard eligibility

User Data Transfer

- One frame type
  - User data
  - No control frame
- No inband signaling
- No sequence numbers
  - No flow nor error control

Required Reading

- Stallings Chapter 11
- ATM Forum Web site
- Frame Relay forum

Cellular Systems-Basic Concepts

- Cellular system solves the problem of spectral congestion.
- Offers high capacity in limited spectrum.
- High capacity is achieved by limiting the coverage area of each BS to a small geographical area called cell.
- Replaces high powered transmitter with several low power transmitters.
- Each BS is allocated a portion of total channels and nearby cells are allocated completely different channels.
- All available channels are allocated to small no of neighboring BS.
- Interference between neighboring BSs is minimized by allocating different channels.

Cellular Systems-Basic Concepts

- Same frequencies are reused by spatially separated BSs.
- Interference between co-channels stations is kept below acceptable level.
- Additional radio capacity is achieved.
- Frequency Reuse-Fix no of channels serve an arbitrarily large no of subscribers

Cellular Concept

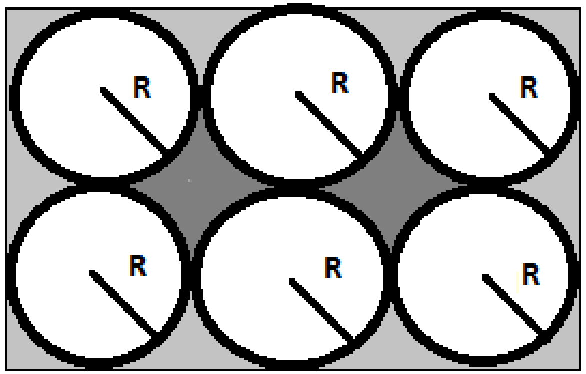
- used by service providers to improve the efficiency of a cellular network and to serve millions of subscribers using a **limited radio spectrum**
- After covering a certain distance a radio wave gets attenuated and the signal falls below a point where it can no longer be used or cause any interference
- A transmitter transmitting in a specific frequency range will have only a limited coverage area
- Beyond this coverage area, that frequency can be reused by another transmitter.
- The entire network coverage area is divided into cells based on the principle of frequency reuse

Frequency Reuse

- The design process of selecting and allocating channel groups for all of the cellular base stations within a system is called frequency reuse or frequency planning.
- Cell labeled with same letter use the same set of frequencies.
- Cell Shapes:
- Circle, Square, Triangle and Hexagon.
- Hexagonal cell shape is conceptual , in reality it is irregular in shape

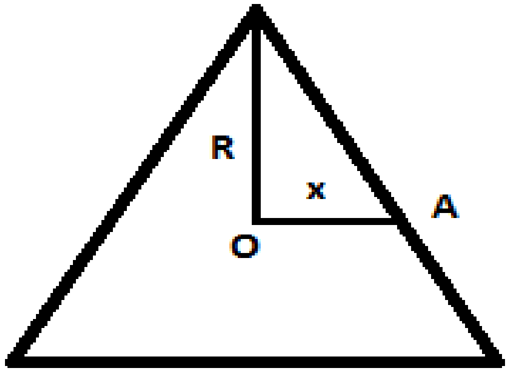
Frequency Reuse

- Cell Geometry :



Circular Cell

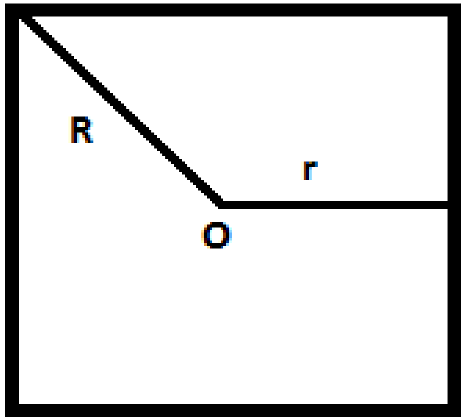
--> Missed Area



Triangular Cell

x < R (Non-Uniform Distribution)

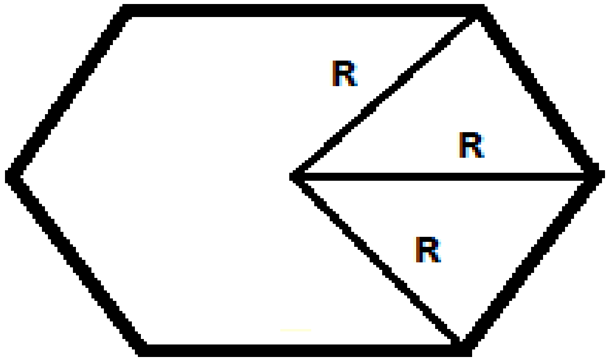
$A_{triangle} = 1.3R^2$



Square cell

r < R (Non-Uniform Distribution)

$A_{Square} = 2.0R^2$

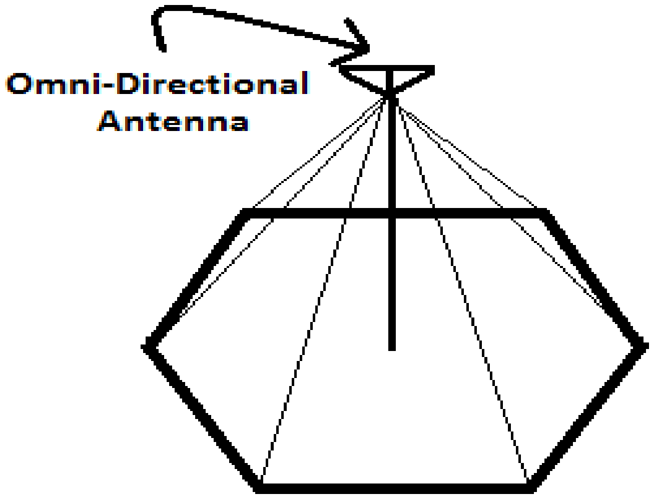


Hexagonal cell
(Uniform Distribution
without overlap &
witout gap area)

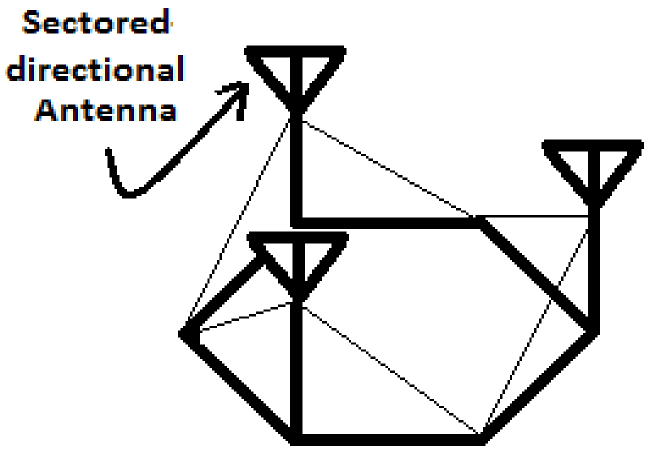
$A_{Hex} = 2.6R^2$

Frequency Reuse

- Hexagonal cell shape is conceptual & is a simplistic model of radio coverage for each base station.
- As the radiation from antenna is circular, thus there is fading on the edges of triangle & Square.
- But in Hexagon shape, there is **lesser fading** on the edges.
- When using hexagonal shaped cell, base station areas are depicted as either being in centre of cell i.e. **Central Excited Cell** or on the 3 edges of the 6 cell vertices i.e. **Edge Excited Cell.**
- For Central Excited Cell, **Omni-directional** antennas are used.
- For Edge Excited Cell, **Sectored-directional** antennas are used.



Centre-Excited cell



Edge-Excited Cell

Frequency Reuse

- The design process of selecting & allocating channel groups for all the cellular base stations with in a system is called **Frequency Reuse** or **Frequency Planning.**
- **Frequency Reuse Distance :** Minimum distance b/w co-channel cells required to keep co-channel interference below certain threshold.
- Each cellular base station is allocated a group of radio channels to be used within a small geographic area called **Cell.**
- Base station of adjacent cells are assigned channel group which contains completely **different channels** than neighboring cells.

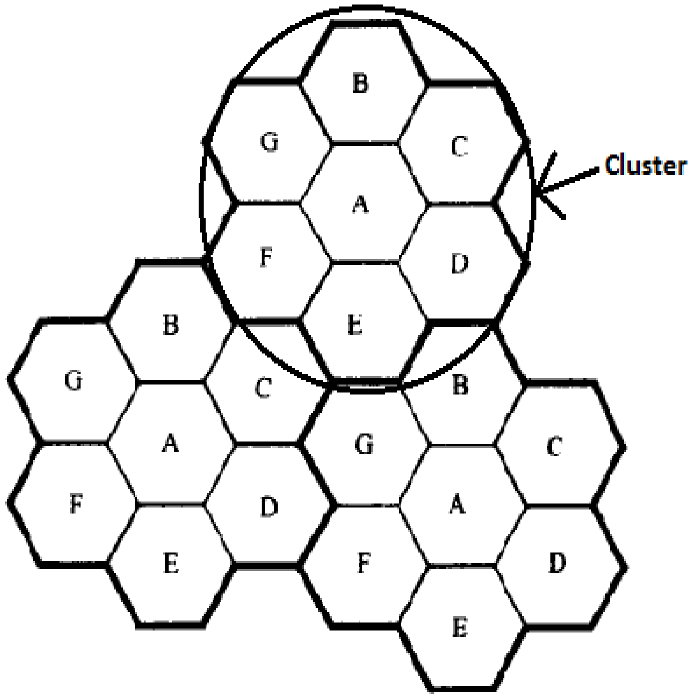


Figure 3.1 Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies. A cell cluster is outlined in bold and replicated over the coverage area. In this example, the cluster size, *N*, is equal to seven, and the frequency reuse factor is 1/7 since each cell contains one-seventh of the total number of available channels.

- If N cells which collectively use the complete set of available frequencies is called a **cluster**.

Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 59.

Locating co-channel Cell

▪To find the nearest co-channel neighbors of a particular cell, one must do the following:

a) Move i cells along any chain of hexagon.
b) Turn 60° counter clockwise & move j cells.

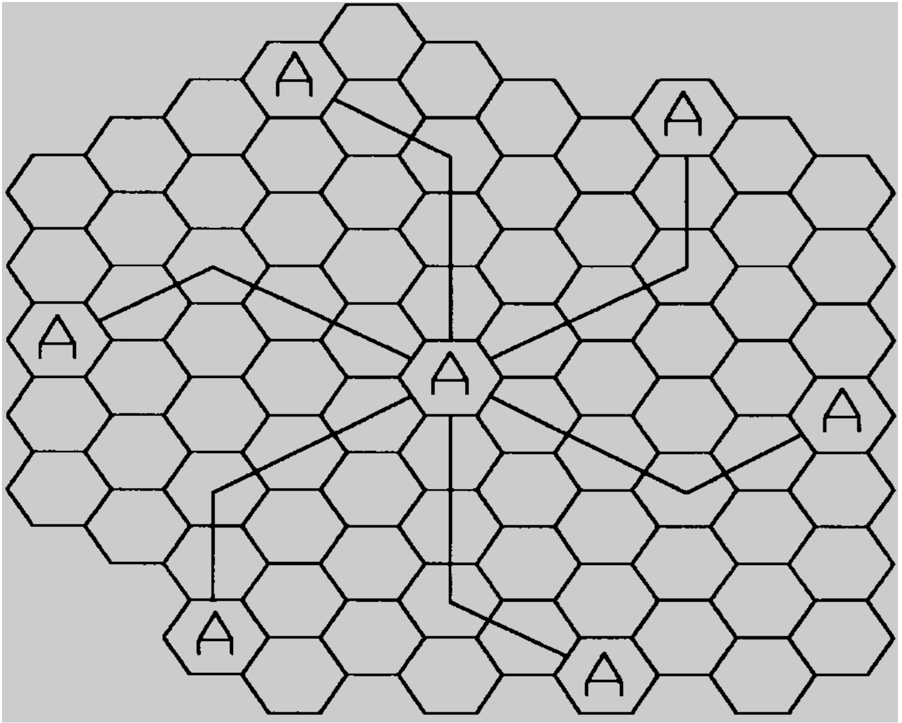
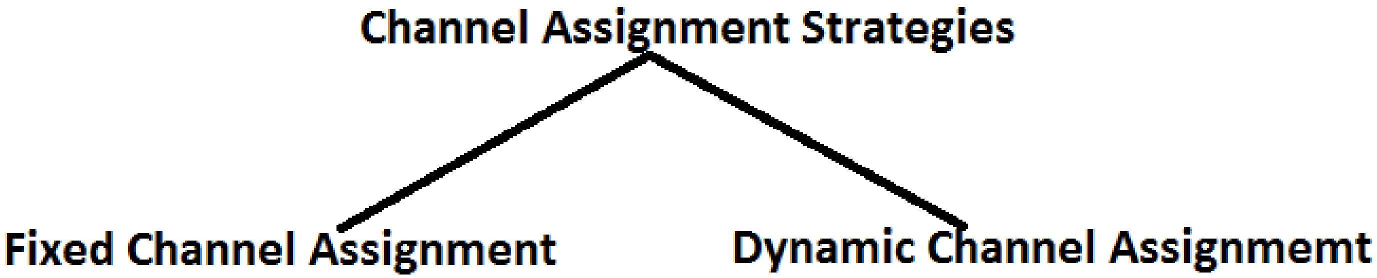


Figure 3.2 Method of locating co-channel cells in a cellular system.
In this example, N = 19 (i.e., i = 3, j = 2). (Adapted from [Oet83] © IEEE.)

Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 60.

Channel Assignment Strategies (CAS)

- Frequency reuse scheme
  - increases capacity
  - minimize interference



- Fixed channel assignment
  - each cell is allocated a predetermined set of voice channel
  - any new call attempt can only be served by the unused channels
  - the call will be *blocked* if all channels in that cell are occupied
- Dynamic channel assignment
  - channels are not allocated to cells permanently.
  - allocate channels based on request.
  - reduce the likelihood of blocking, increase capacity.

Fixed Channel Assignment Strategy (FCAS)

- In FCAS each cell is assigned a *predetermined* set of voice channels
- Any call attempt within the cell can only be served by the *unused* channel in that particular cell
- If all the channels in the cell are occupied, the call is *blocked*. The user does not get service.
- In variation of FCA, a cell can *borrow channels* from its neighboring cell if its own channels are full.

Dynamic Channel Assignment (DCA)

- Voice channels are not allocated to different cells *permanently*.
- Each time a call request is made, the *BS request* a channel from the MSC.
- MSC allocates a channel to the requesting cell using an algorithm that takes into account
  - likelihood of future blocking
  - The reuse distance of the channel (should not cause interference)
  - Other parameters like cost
- DCA reduce the likelihood of blocking and increases capacity
- Requires the MSC to collect realtime data on channel occupancy and traffic distribution on continous basis.
- DCA is more complex (real time), but reduces likelihood of blocking

Handoff

- When a mobile moves into a different cell while conversation is in progress, the MSC automatically transfers the call to a new channel belonging to new base station is known as **Handoff.**
- Handoff operation not only involves **identifying a new base station**, but also requires that the **voice & control signals** be allocated to channels with the new base station.
- Handoff must be **performed successfully** & as **infrequently** as possible.
- For this system designer must specify an **optimum signal level** at which to **initiate** a handoff.
- Once a particular signal level is specified as the **minimum usable signal** for acceptable voice quality at the base station receiver, a slightly stronger signal level is used as a **threshold** at which a handoff is made.
- This margin (Threshold) is given by:

$$\Delta = P_r(\text{HANDOFF}) - P_r(\text{MIN. USABLE})$$

Handoff

- Δ can't be too large or too small.

Case 1: if Δ is **too large**, unnecessary handoffs are there, which burden the MSC.

Case 2: if Δ is **too small**, there may be insufficient time to complete a handoff before a call is lost due to weak signal conditions.

- Call Drop event can happen:

Case 1: When there is an excessive delay by the MSC in assigning a handoff.

Excessive delay may occur during:

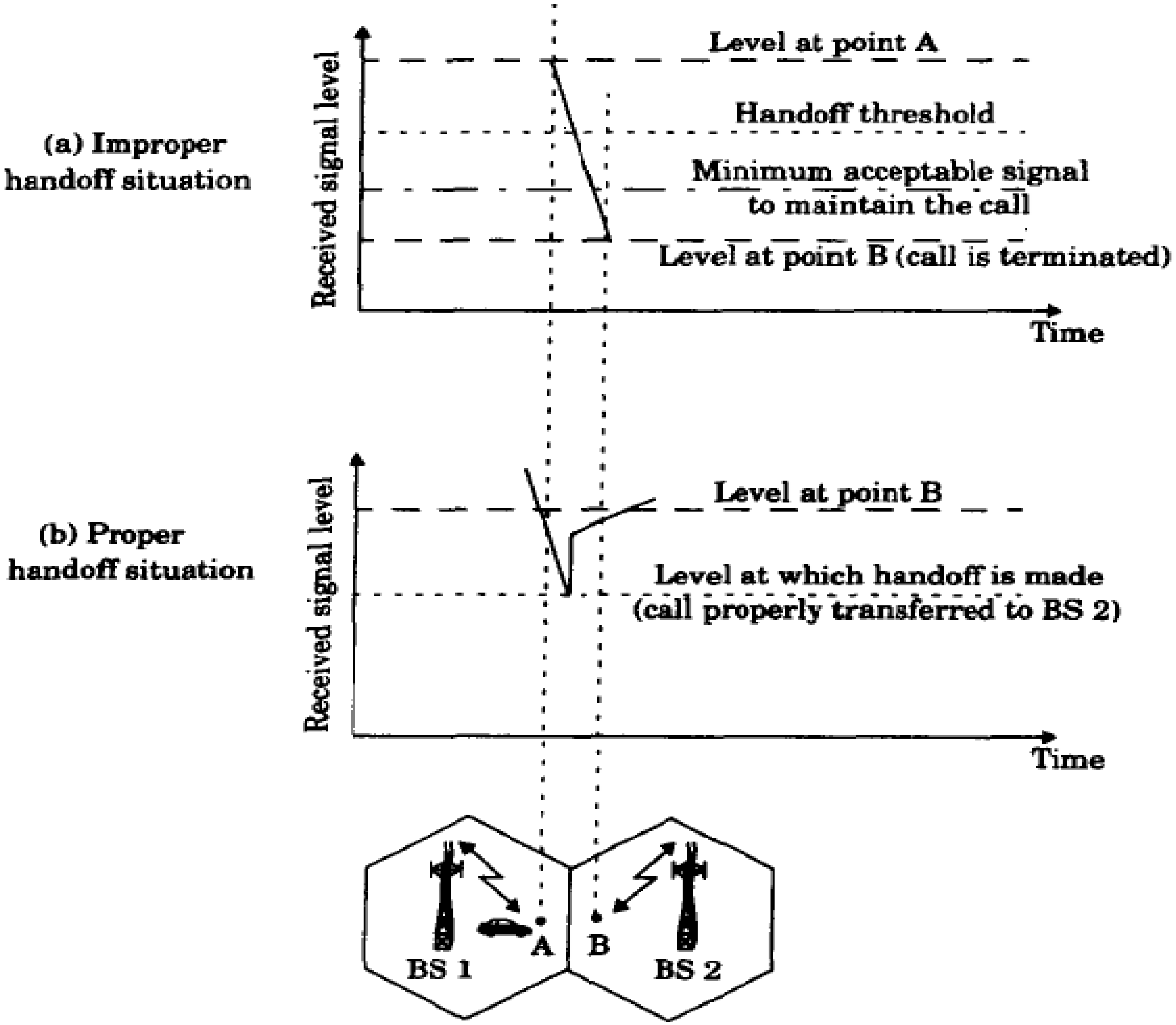
1) High traffic density, due to computational loading at the MSC

2) Due to the fact that no channels are available on any of the nearby base stations.

Case 2: When the threshold Δ is set too small for the handoff time in the system.

- **Dwell Time :** The time over which a call be maintained within a cell without handoff is called **"Dwell Time".** (after the handoff threshold)

Handoff

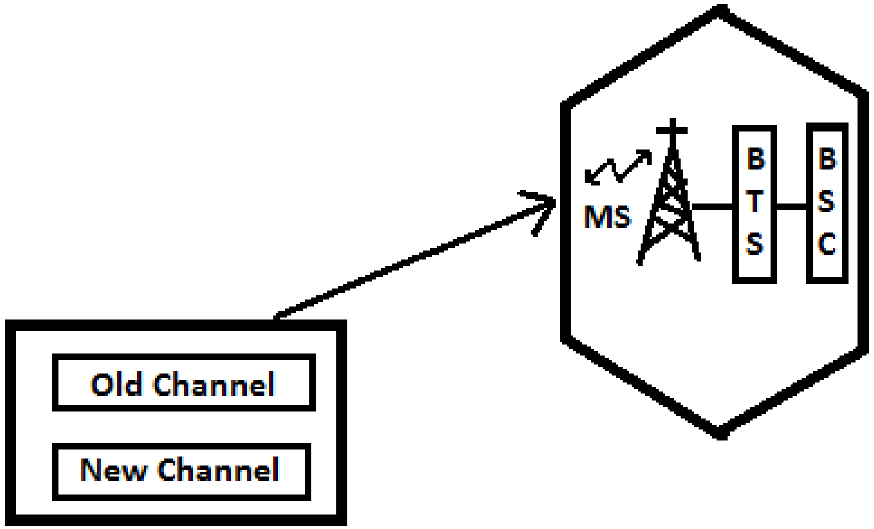


Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 63.

Types of Handoff

1) **Intra-cell-Intra BSC Handover**
2) **Inter-cell-Intra BSC Handover**
3) **Inter-cell-Inter BSC Handover**
4) **Inter MSC Handover**

1) **Intra-cell-Intra BSC Handover :** Smallest of the Handover is the intra-cell handover where the subscriber is handed over to another traffic channel (generally in another frequency) with in same cell.

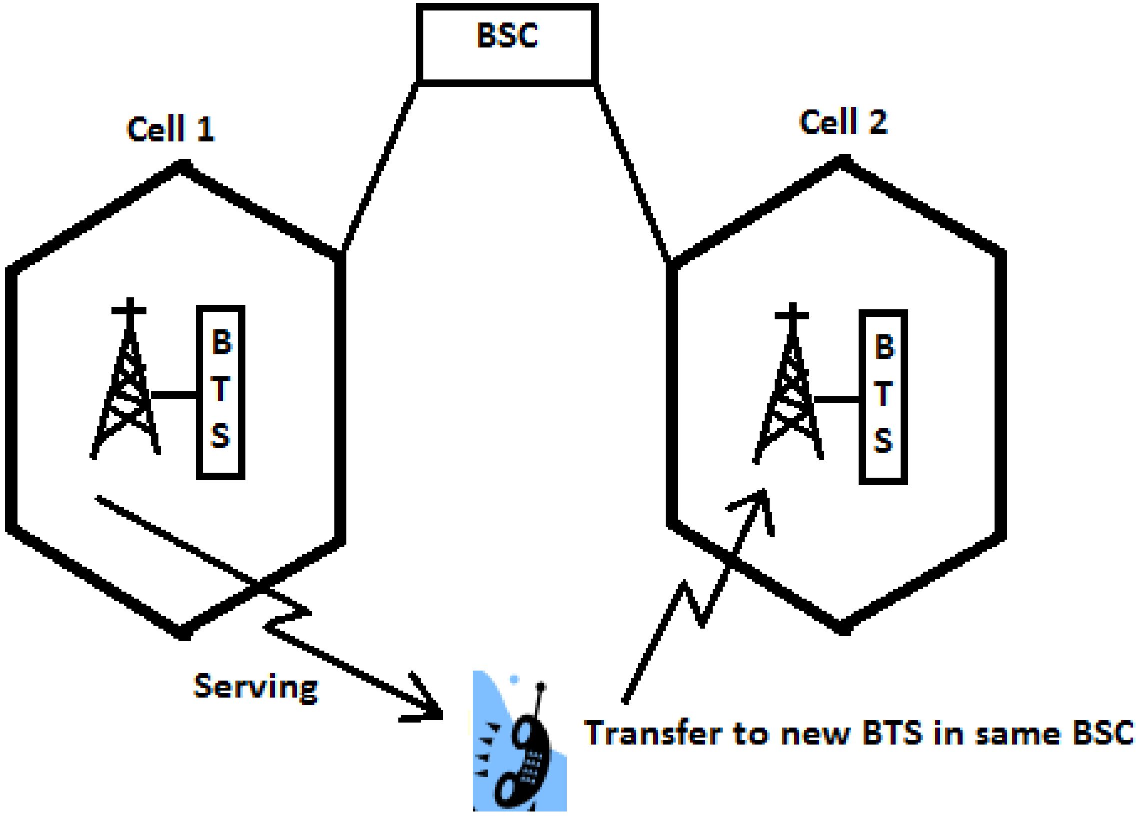


- In this case, BSC controlling the cell makes the decision to perform handover.

Types of Handoff

2) Inter-cell-Intra BSC Handover :

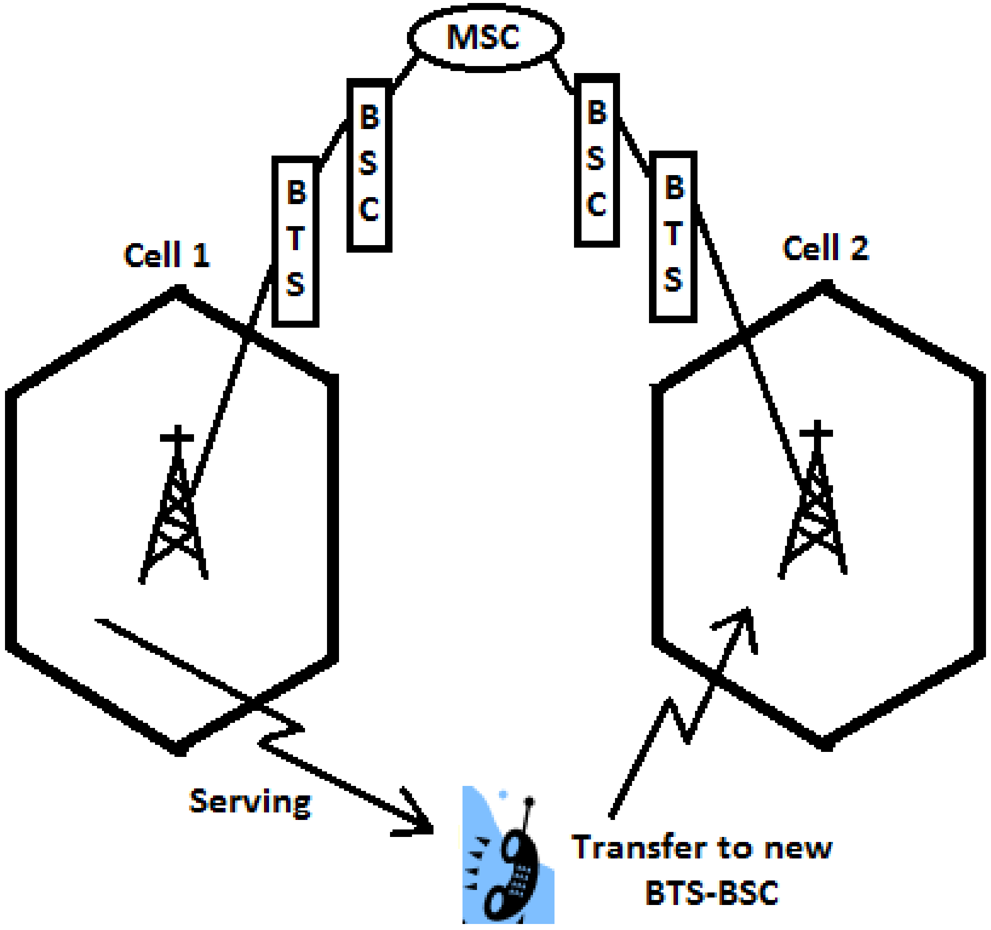
- The subscriber moves from cell1 to cell 2 but within BSC.
- In this case, the handover process is carried out by the BSC.
- Traffic connection with cell 1 is released when the connection with cell 2 is setup successfully.



Types of Handoff

3) Inter-cell-Inter BSC Handover :

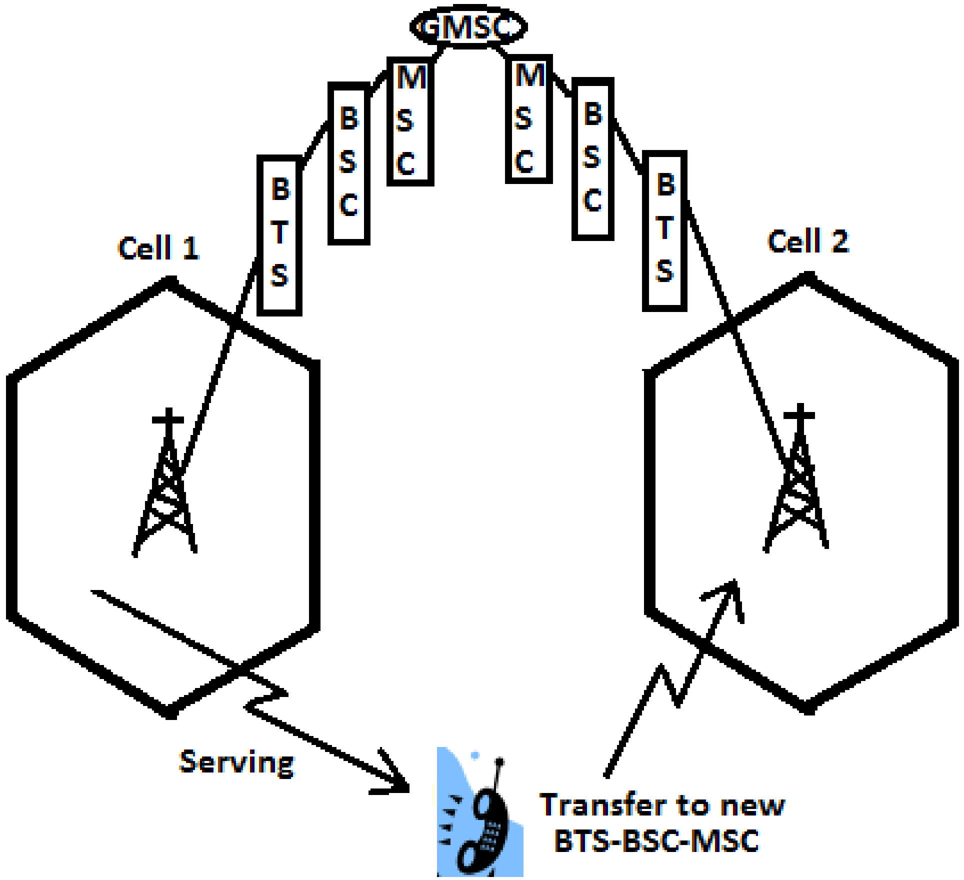
- The subscriber moves from cell1 to cell 2 which is served by another BSC.
- In this case, handover process is carried out by the MSC, but the decision to make the handover is still done by the first BSC.
- Traffic connection with the first BSC & BTS is released when the connection with the new BSC & BTS is setup successfully.



Types of Handoff

4) Inter MSC Handover :

- The subscriber moves from cell1 to cell 2 which is served by another MSC.
- In this case, handover process is carried out by the GMSC.
- Traffic connection with the first BTS-BSC-MSC is released when the connection with the new BTS-BSC-MSC is setup successfully.



Classification of Handoff

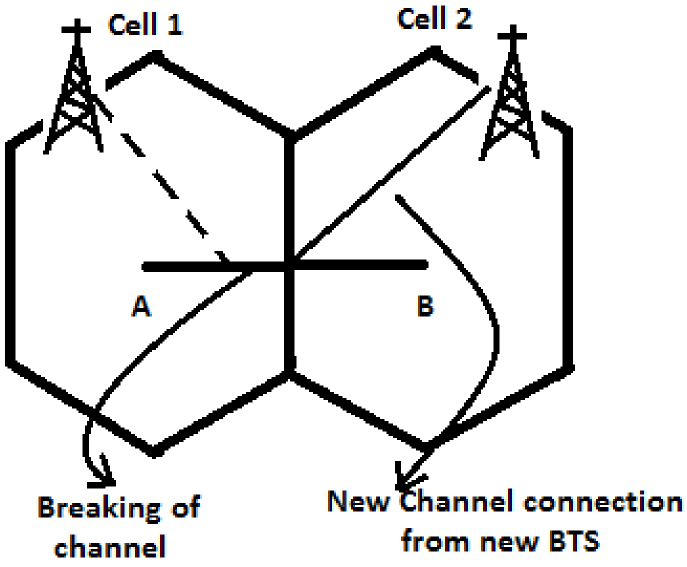
- Handoff can be classified as:

1) Hard Handoff

2) Soft Handoff

1) <u>Hard Handoff</u> :

- Also known as **Break Before Make.**
- It is one in which the channel in the source cell is released & only then the channel in the target cell is engaged.
- Thus the connection to the source is broken before the connection to the target is made.
- Hard handoffs are intended to be **instantaneous** in order to minimize the disruption to the call.
- It is not necessary that there is always a connection b/w base station & mobile station.

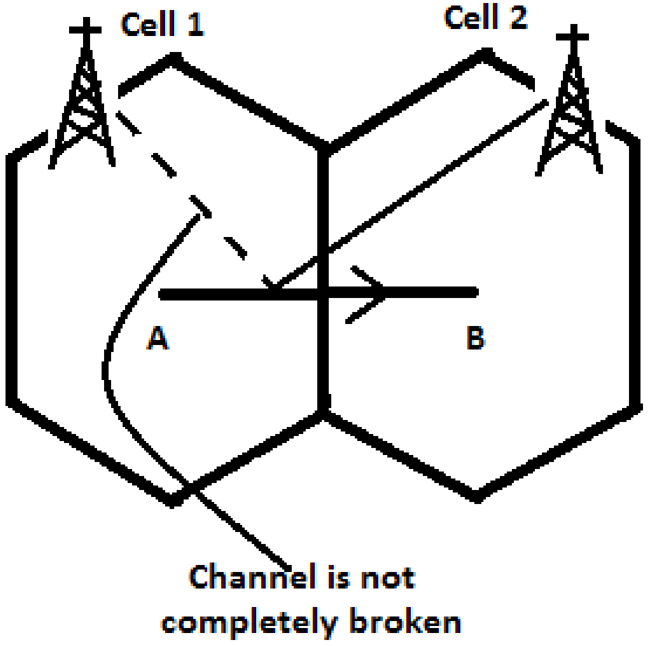


Classification of Handoff

2) Soft Handoff :

- Also known as **Make Before Break.**

  It is one in which the channel in the source cell is retained & used for a while in parallel with the channel in target cell.
- Thus the connection to the target is established before the connection to the source is broken.
- In this kind of handoff, there is always a connection b/w base station & mobile station.



Prioritizing Handoff / Guard Channel Concept

- Here some of the channels of total available channels is **reserved** for handoff request from on-going calls which may be handed-off into the cell.
- Guard channels however offer **efficient spectrum utilization** when dynamic channel assignment strategies, which minimize the number of required guard channels by efficient demand-based allocation are used.

Queuing of handoff request :

- It is another method to **decrease** the probability of **forced termination of calls** due to lack of available channels.
- Queuing of handoffs is possible due to the fact there is a finite time interval b/w the time the received signal level drops below the handoff threshold & the time the call is terminated due to insufficient signal level.

Practical Handoff Considerations

- Problems occur because of a *large range of mobile velocities*
  - pedestrian vs. vehicle user
  - A pedestrian never requires handoff during the call, but if high speed user passes constantly b/w very small cells, MSC gets burdened quickly.
- Small cell sizes and/or micro-cells → *larger # handoffs*
- MSC load is *heavy* when high speed users are passed between very small cells
- **Umbrella Cells**
  - use *different antenna heights* and *Tx power levels* to provide large **and** small cell coverage
  - multiple antennas & Tx can be co-located at single location if necessary (saves on obtaining new tower licenses)
  - large cell → high speed traffic → fewer handoffs
  - small cell → low speed traffic
  - example areas: interstate highway passing through urban center, office park, or nearby shopping mall

Umbrella Cells

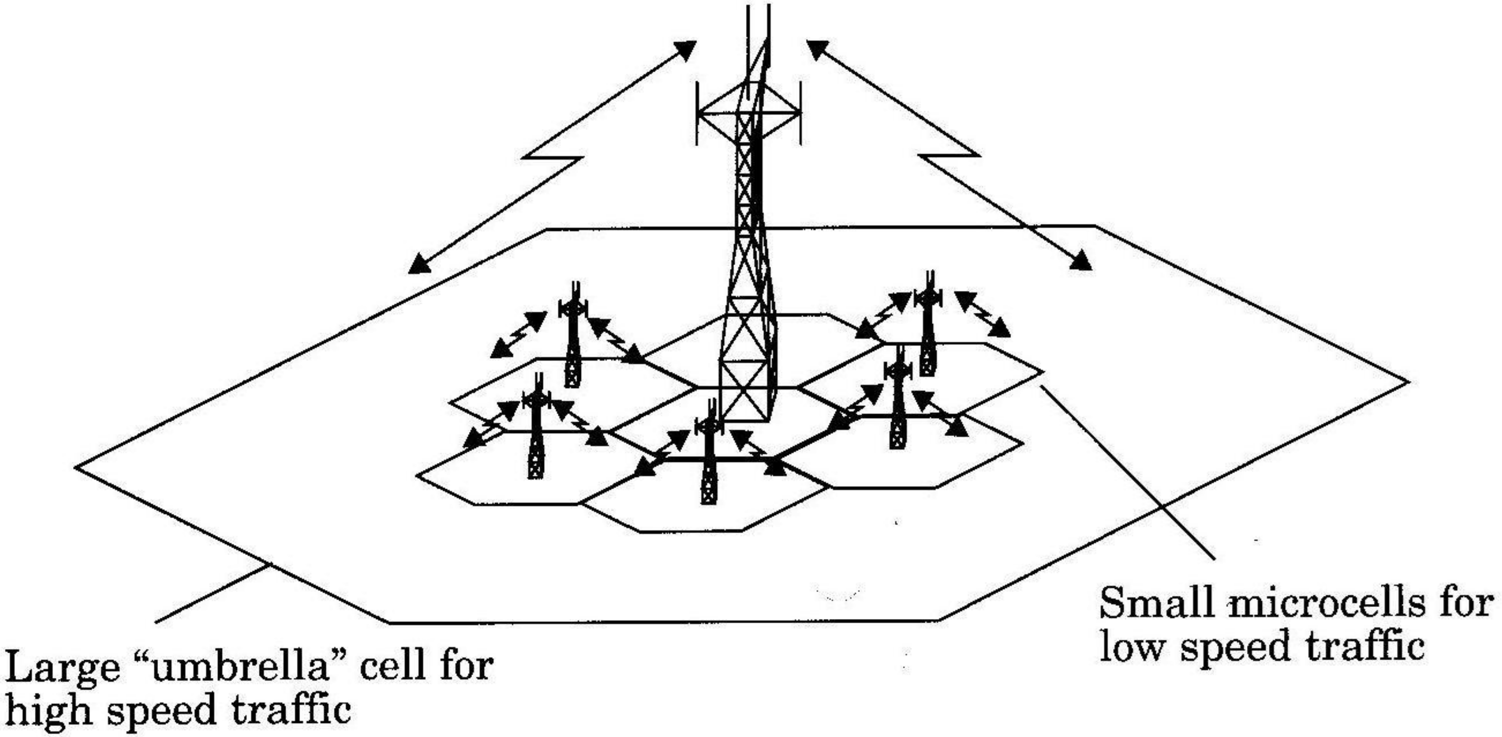


Figure 3.4 The umbrella cell approach.

Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 67.

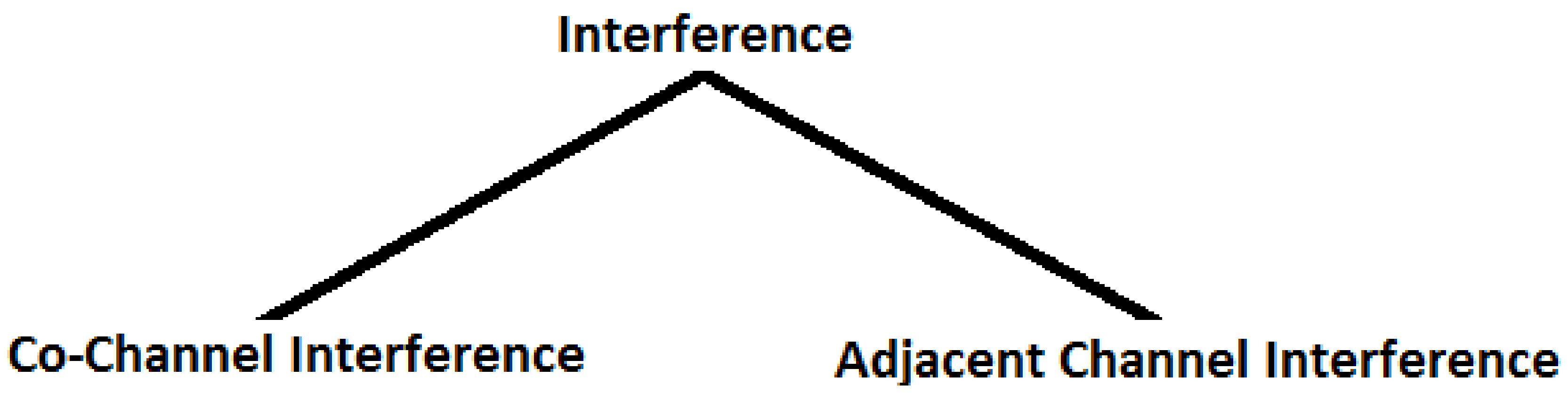
Typical handoff parameters

- Analog cellular (1st generation)
  - threshold margin △ ≈ 6 to 12 dB
  - total time to complete handoff ≈ 8 to 10 sec

- Digital cellular (2nd generation)
  - total time to complete handoff ≈ 1 to 2 sec
  - lower necessary threshold margin △ ≈ 0 to 6 dB
  - enabled by mobile assisted handoff

Interference & System Capacity

- **Source of interference :**
- Another mobile in the same cell.
- A call in progress in a neighboring cell.
- Other base stations operating in the same frequency band.
- Any non-cellular system which inadvertently leaks energy into cellular frequency band.



Effects of Interference

- Interference in voice channels causes
  - Crosstalk
  - Noise in background
- Interference in control channels causes
  - Error in digital signaling, which causes
    - Missed calls
    - Blocked calls
    - Dropped calls

Interference & System Capacity

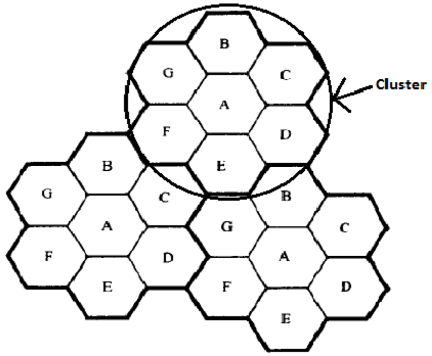
Co-Channel Interference & System Capacity :

- Frequency reuse implies that in a given coverage area there are several cells that use the same set of frequencies. These cells are called **Co-Channel cells** & interference b/w signals from these cells is called **Co-Channel Interference.**
- Unlike, **thermal noise** which can be overcome by increasing the SNR (signal-to-noise ratio), Co-Channel interference can not be combated by simply increasing the carrier power of a transmitter, because an **increase in carrier transmit power** increases the **interference to neighboring co-channel cells**.
- **To reduce co-channel interference**, co-channel cells must be physically separated by a **minimum distance** to provide sufficient isolation due to propagation.

R **-** Radius of a cell

D - distance b/w centers of the nearest

co-channel cells.



Therefore by **increasing the ratio D/R**, separation b/w co-channel cells relative to the coverage distance of a cell is increased. Thus **interference is reduced**.

Interference & System Capacity

- **Q Co-Channel Reuse ratio**, is related to cluster size shown in table. For Hexagonal geometry

$$Q = \frac{D}{R} = \sqrt{3N}$$

Table Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 69.

Co-channel Reuse Ratio for Some Values of N

	Cluster Size (N)	Co-channel Reuse Ratio(Q)
$i = 1, j = 1$	3	3
$i = 1, j = 2$	7	4.58
$i = 2, j = 2$	12	6
$i = 1, j = 3$	13	6.24

- A **small value of Q** provides **larger capacity,** Since the cluster size N is small.
- A **large value of Q** improves the **Transmission Quality,** Due to smaller level of co-channel interference.

Therefore a **Trade-off** must be made b/w these two objectives in actual cellular design.

Interference & System Capacity

- Let i_o be the number of co-channel interfering cells.

Then, the signal-to-interference ratio (S/I or SIR) for a mobile receiver which monitors a forward channel can be expressed as:

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{i_0} I_i}$$

where S is the desired signal power from the desired base station & I is the interference power caused by the i^{th} interfering co-channel cell base station.

- Average Power Received P_r at a distance d from the transmitting antenna is approximated by:

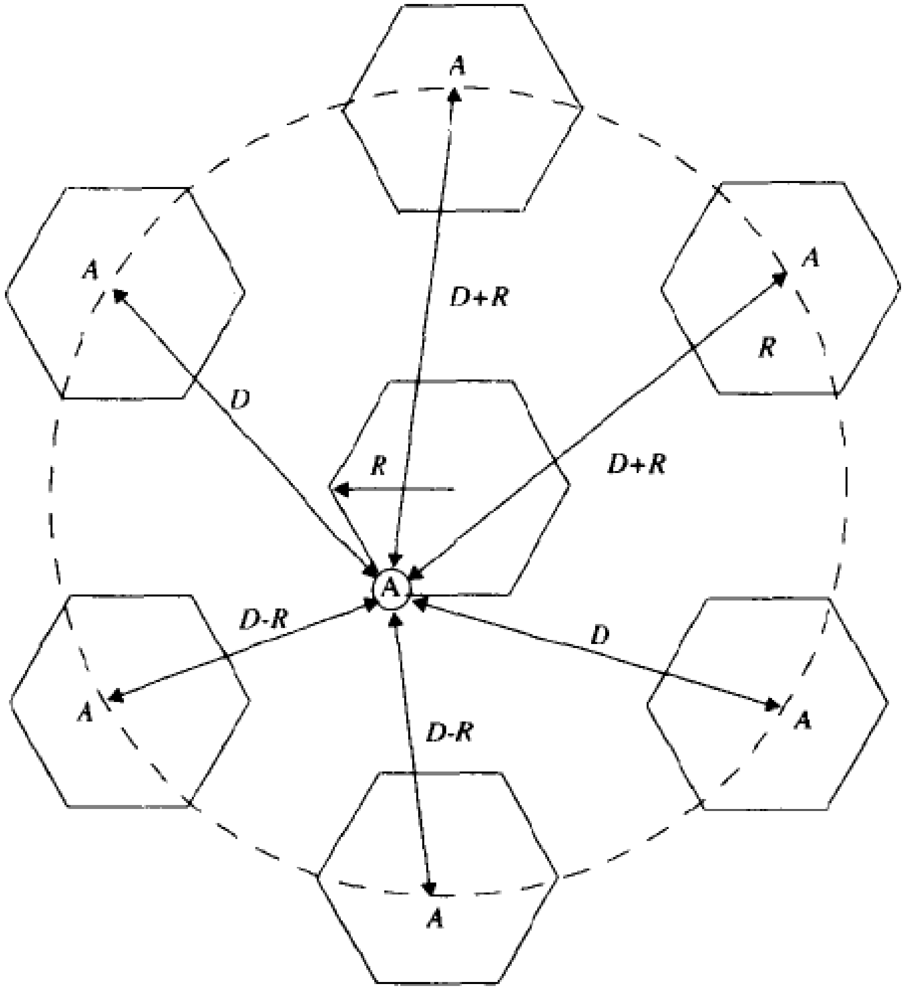
$$P_r = P_0\left(\frac{d}{d_0}\right)^{-n}$$

$$P_r\,(\mathrm{dBm}) = P_0\,(\mathrm{dBm}) - 10n\log\left(\frac{d}{d_0}\right)$$

where $\mathbf{P_o}$ is the power received at a close-in reference point in the far field region of the antenna at a small distance $\mathbf{d_o}$ from the transmitting antenna & **n** is the path loss exponent.

Interference & System Capacity

- Now consider the **forward link** where the desired signal is the serving base station & where the interference is due to co-channel base station.
- If D_i is the distance of the i^{th} interfere from the mobile, the received power at a given mobile due to the i^{th} interfering cell will be proportional to $(Di)^{-n}$.
- When transmit power of each base station is equal & the path Loss exponent is the same throughout the coverage area, S/I for a mobile can be approximated as:

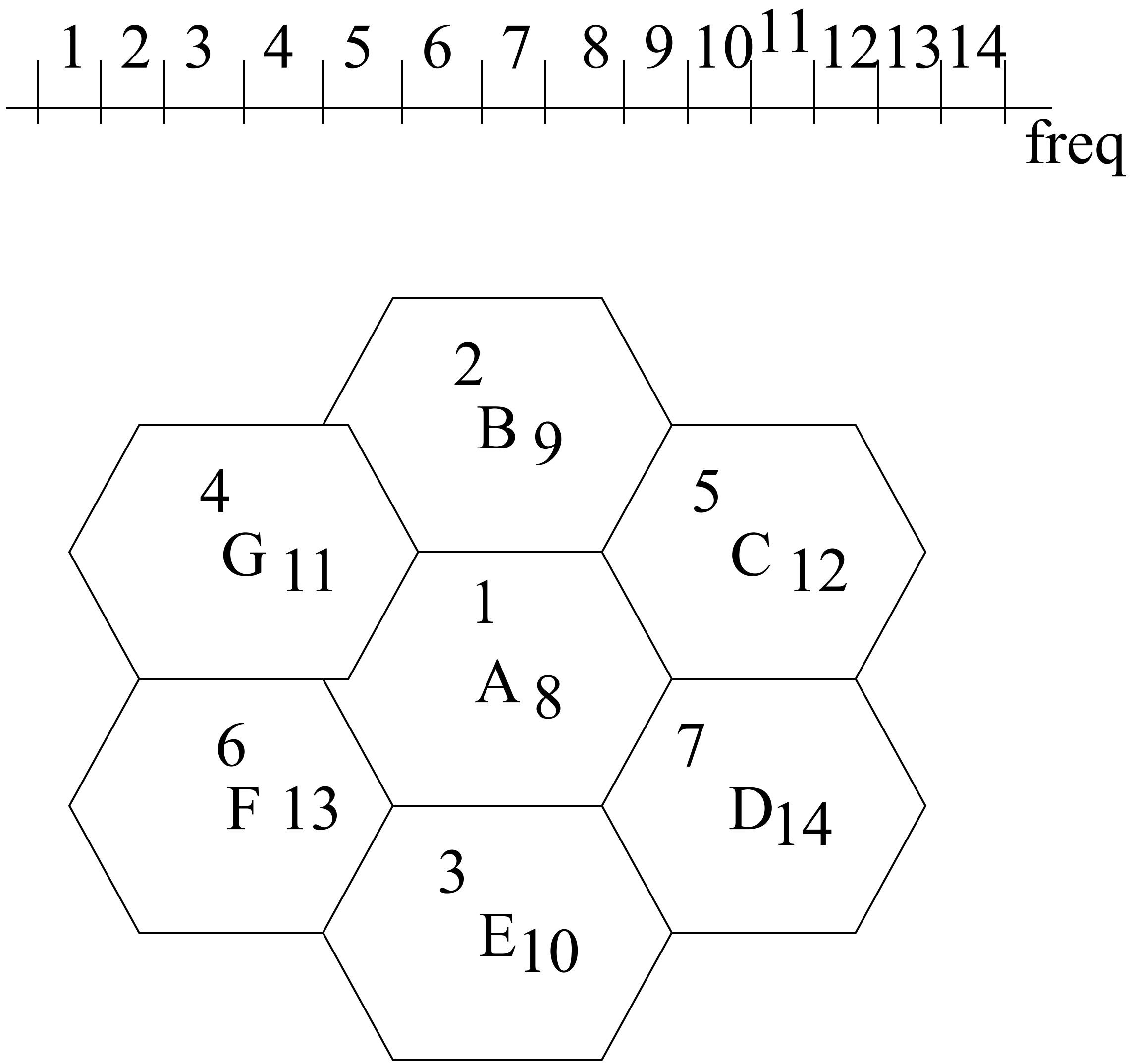


$$\frac{S}{I} = \frac{R^{-n}}{\sum_{i=1}^{i_0} (D_i)^{-n}} \quad \text{---------(A)}$$

Source : Wireless Communications: Principles and Practice, Theodore S. Rappaport, pp 71.

Adjacent Channel Interference

- Results from imperfect receiver filters, allowing nearby frequencies to leak into pass-band.
- Can be minimized by careful filtering and channel assignments.
- Channels are assigned such that frequency separations between channels are maximized.
- For example, by sequentially assigning adjacent bands to different cells
- Total 832 channels, divided into two groups with 416 channels each.
- Out of 416, 395 are voice and 21 are control channels.
- 395 channels are divided into 21 subsets, each containing almost 19 channels, with closet channel 21 channels away
- If N=7 is used, each cell uses 3 subsets, assigned in such a way that each channel in a cell is 7 channels away.



Frequency Planning/Channel Assignment

Multiple Access Techniques

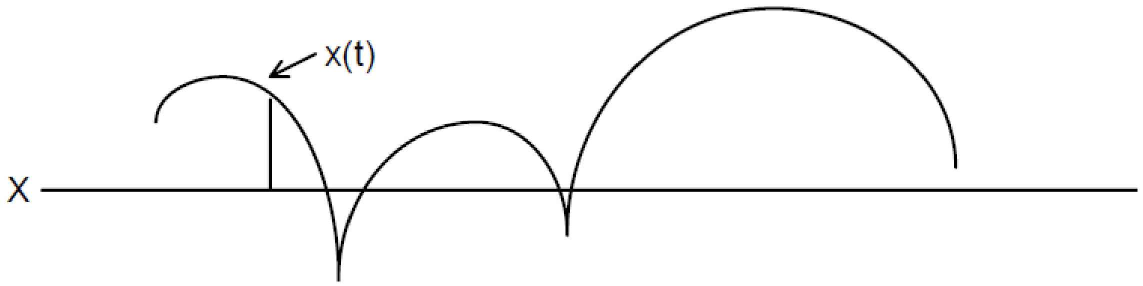
- The transmission from the BS in the downlink can be heard by each and every mobile user in the cell, and is referred as *broadcasting*. Transmission from the mobile users in the uplink to the BS is many-to-one, and is referred to as multiple access.
- Multiple access schemes to allow many users to share simultaneously a finite amount of radio spectrum resources.
  - Should not result in severe degradation in the performance of the system as compared to a single user scenario.
  - Approaches can be broadly grouped into two categories: narrowband and wideband.

Multiple Access Techniques

- Multiple Accessing Techniques : with possible conflict and conflict- free
  - Random access
  - Frequency division multiple access (FDMA)
  - Time division multiple access (TDMA)
  - Spread spectrum multiple access (SSMA) : an example is Code division multiple access (CDMA)
  - Space division multiple access (SDMA)

Multiple Access Techniques -- Narrowband Systems

- Transmission experiences nonselective fading. This means that when fades occur, all of the information (i.e. the whole channel) is affected.
- Channel system : generally total spectrum is divided into a number of relatively narrow radio channels (e.g. FDMA). Occurrence of call blocking if channels are all being used. Unused bandwidth in each channel cannot be used by other users.



Narrowband Systems (Interleaving + coding techniques are used)

- Data generated in source 1 2 3 4 5 … 8 9 10 11 12 … 15 16 17 18 …
- Transmitted data 1 8 15 22 29 … 2 9 16 23 … 3 10 17 24 …
- Received data 1 8 15 22 29 … 2 9 16 23 30 … 3 10 17 24 31 …
- with the shaded data bits undergo deep fade
- Deinterleaved data 1 2 3 4 5 … 8 9 … 15 16 17 18 …

1	8	15	22	29	…..	
2	9	16	23	30	…..	
3	10	17	24	31	……	
4	11	18	25	32	……	
5	12	19	26	33	……	
6	13	20	27	34	……	
7	14	21	28	35	……	

Multiple Access Techniques -- Wideband Systems

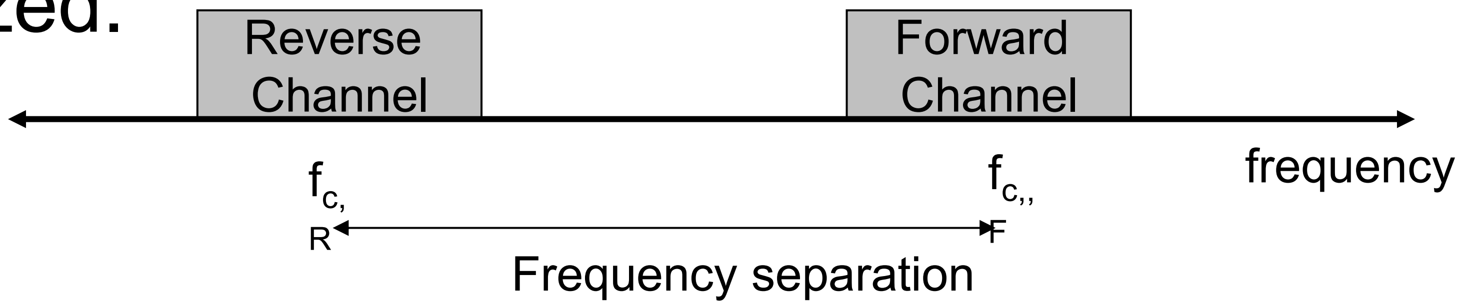
- The main feature of wideband systems is that either all the spectrum available (e.g. CDMA, TDMA) or a considerable portion of it is used by each user (e.g. TDMA+FDMA).
- The advantage of wideband systems is that the transmission bandwidth always exceeds the coherence bandwidth for which the signal experiences only selective fading. That is, only a small fraction of the frequencies composing the signal is affected by fading.
- Signal can be distorted and therefore equalization is needed but unlikely that a total signal fade will occur.

Duplexing

- For voice or data communications, must assure two way communication (duplexing, it is possible to talk and listen simultaneously). Duplexing may be done using frequency or time domain techniques.
  - Forward (downlink) band provides traffic from the BS to the mobile
  - Reverse (uplink) band provides traffic from the mobile to the BS.

Frequency division duplexing (FDD)

- Provides two distinct bands of frequencies for every user, one for downlink and one for uplink.
- A large interval between these frequency bands must be allowed so that interference is minimized.



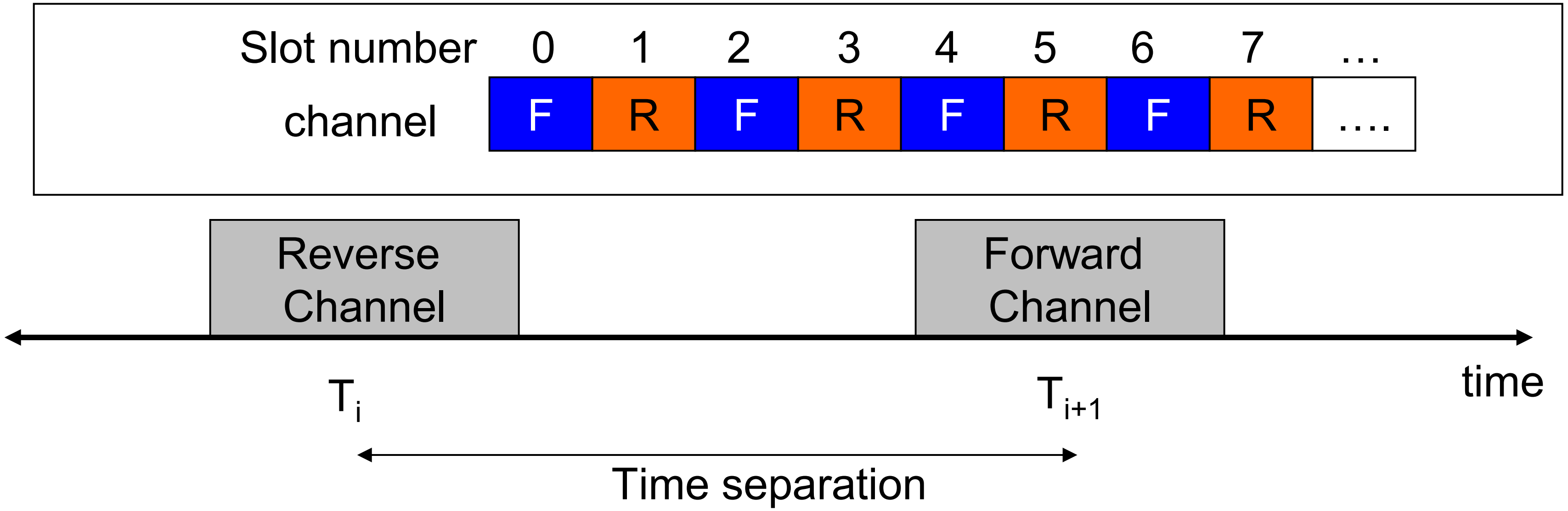
Frequency separation should be carefully decided

Frequency separation is constant

Time division duplexing (TDD)

- In TDD communications, both directions of transmission use one contiguous frequency allocation, but two separate time slots to provide both a forward and reverse link.
- Because transmission from mobile to BS and from BS to mobile alternates in time, this scheme is also known as “ping pong”.
- As a consequence of the use of the same frequency band, the communication quality in both directions is the same. This is different from FDD.

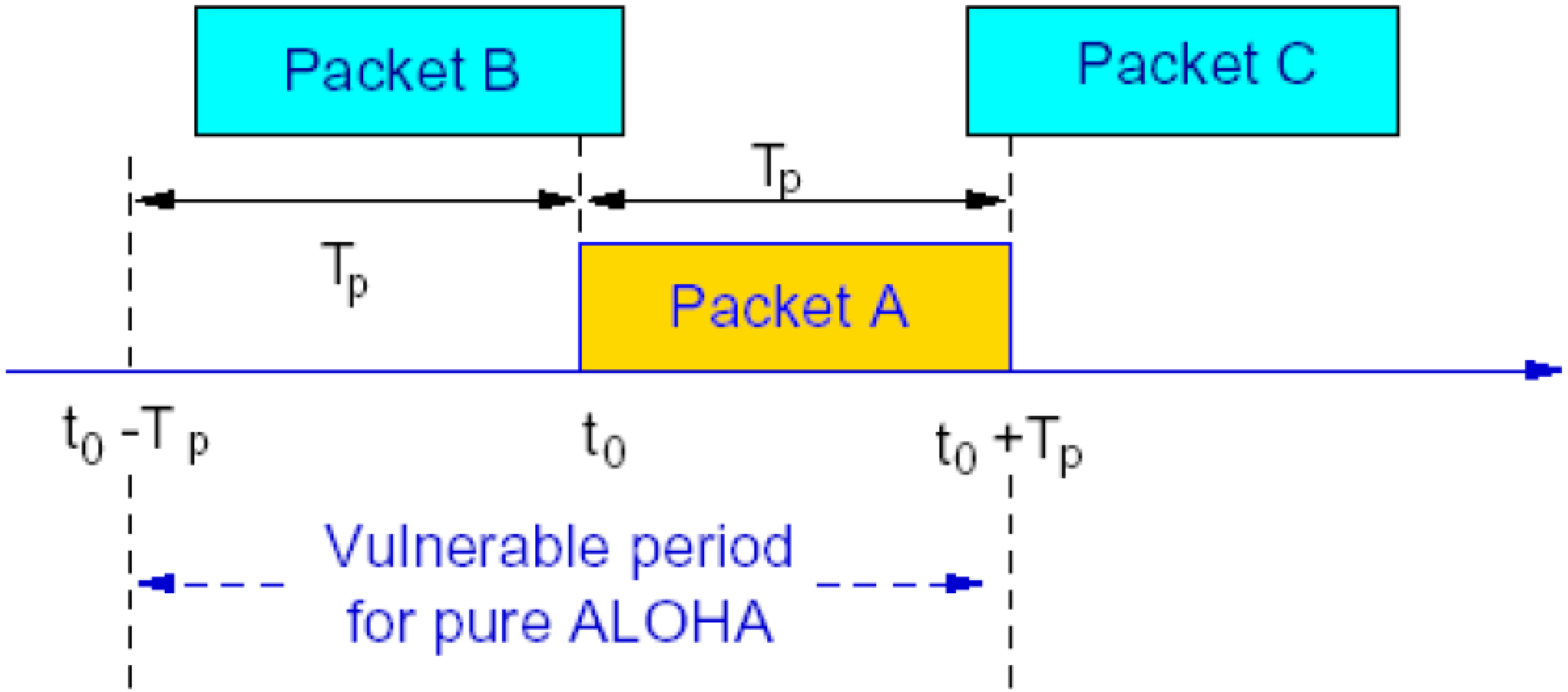
Time division duplexing (TDD)



Random Access -- Aloha systems

- Aloha is a packet-switching system. The time interval required to transmit one packet is called a *slot*.
- When transmissions from two or more users overlap, they destroy each other, whether it is complete overlap or partial overlap – collision takes place.
- The maximum interval over which two packets can overlap and destroy each other is called the *vulnerable* period.
- The mode of random access in which users can transmit at anytime is called *pure Aloha.* In this case, the vulnerable period is two slot times.
- A version in which users are restricted to transmit only from the instant corresponding to the slot boundary is referred to as *slotted Aloha*. The alignment of transmission to coincide with the slot boundary means that packets can only experience complete overlap, so the vulnerable period is only one slot time.

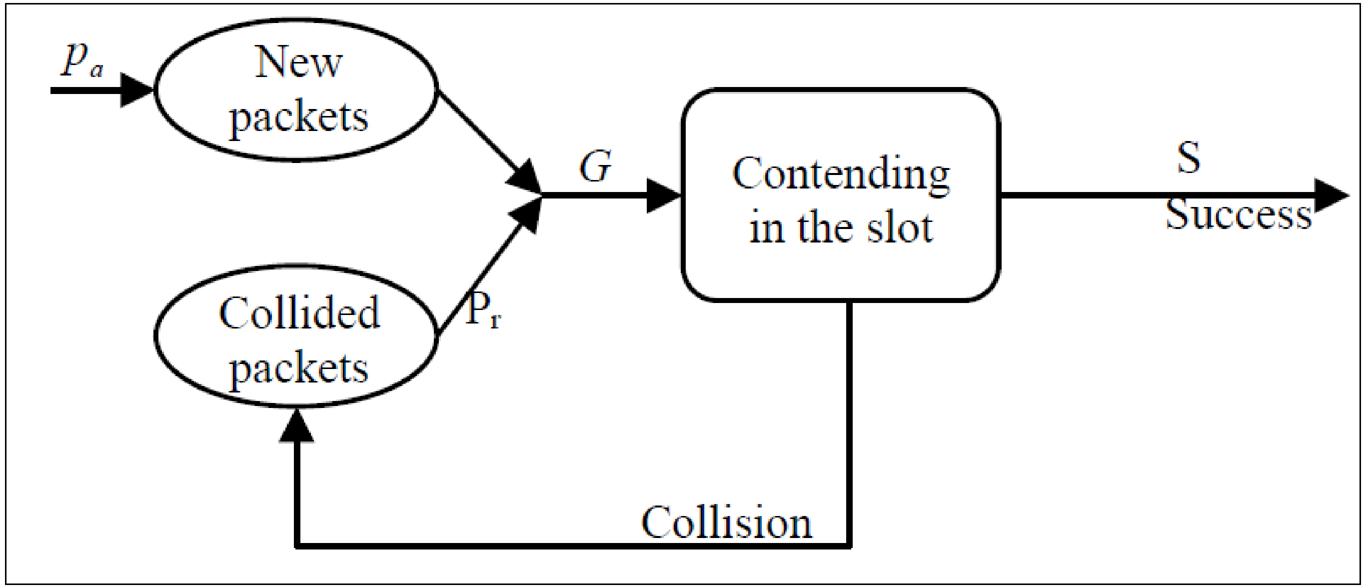
Random Access -- Aloha systems



Throughput of Aloha systems

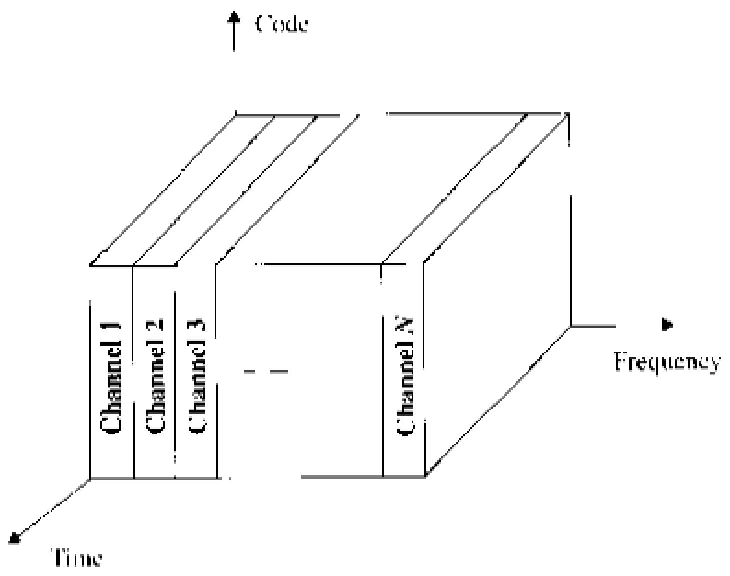
- In the Aloha systems, a transmitting user upon hearing a collision, backs off for a random delay interval and transmits again, until success is achieved. This is known as collision resolution.
- We assume that the total traffic generated for transmission (including re-transmission) obeys a Poisson distribution.
- The transmission is successful when there is no other packet transmission during the vulnerable period.

Throughput of Aloha systems



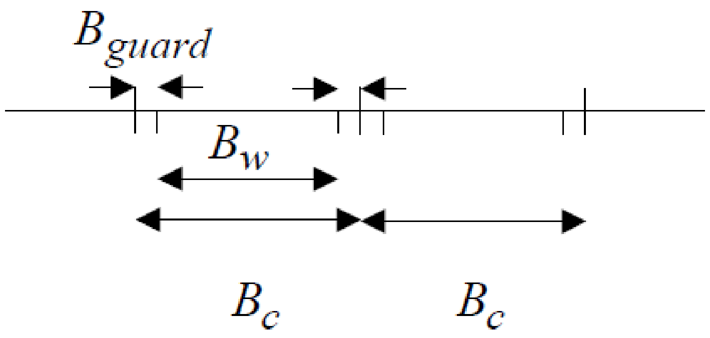
FDMA

- In FDMA, each user is allocated a unique frequency band or channel. During the period of the call, no other user can share the same frequency band.



FDMA

- All channels in a cell are available to all the mobiles. Channel assignment is carried out on a first-come first- served basis.
- The number of channels, given a frequency spectrum *BT* , depends on the modulation technique (hence *Bw* or *Bc*) and the guard bands between the channels 2*Bguard* . These guard bands allow for imperfect filters and oscillators and can be used to minimize adjacent channel interference.
- FDMA is usually implemented in narrowband systems.

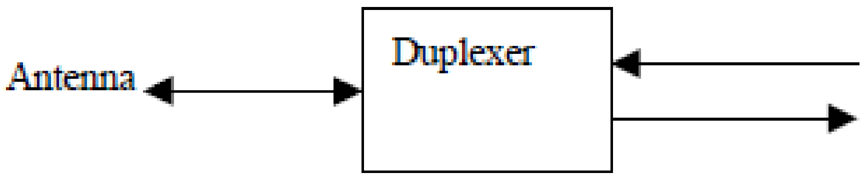


Main features

- Continuous transmission : the channels, once assigned, are used on a non-time-sharing basis. This means that both subscriber and BS can use their corresponding allotted channels continuously and simultaneously.
- Narrow bandwidth : Analog cellular systems use 25-30 kHz. Digital FDMA systems can make use of low bit rate speech coding techniques to reduce the channel band even more.
- If FDMA channels are not in use, then they sit idle and cannot be used by other users to increase capacity.
- Low ISI : Symbol time is large compared to delay spread. No equalizer is required (Delay spread is generally less than a few μs – flat fading).

Main features

- Low overhead : Carry overhead messages for control, synchronization purposes. As the allotted channels can be used continuously, fewer bits need to be dedicated compared to TDMA channels.
- Simple hardware at mobile unit and BS : (1) no digital processing needed to combat ISI (2) ease of framing and synchronization.
- Use of duplexer since both the transmitter and receiver operate at the same time. This results in an increase in the cost of mobile and BSs.

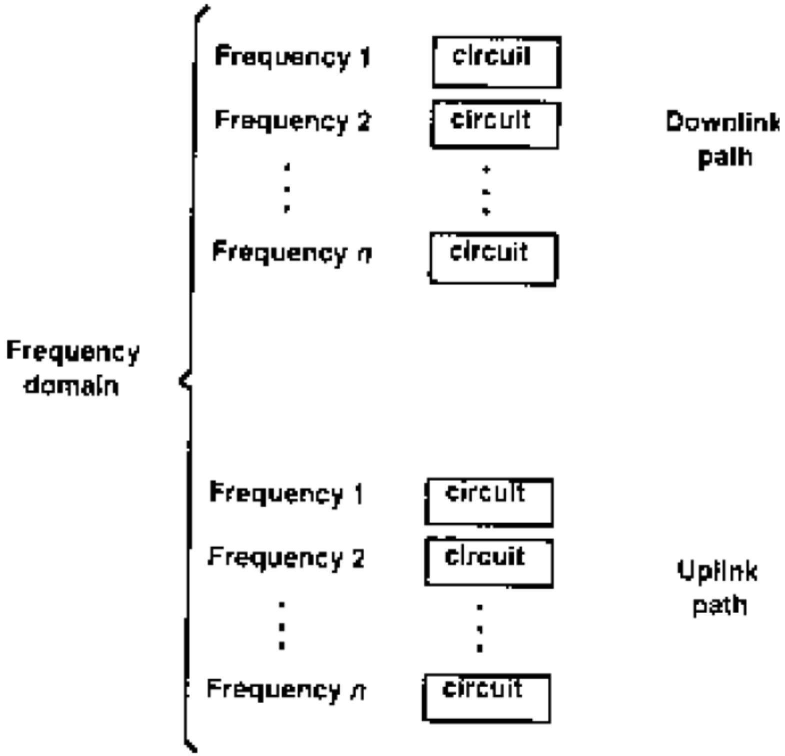


- FDMA required tight RF filtering to minimize adjacent channel interference.

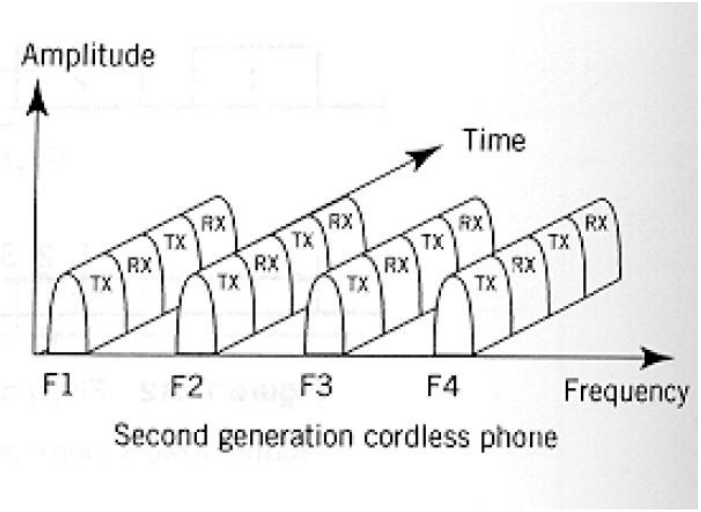
Nonlinear effects in FDMA

- In a FDMA system, many channels share the same antenna at the BS. The power amplifiers or the power combiners, when operated at or near saturation are nonlinear.
- The nonlinear ties generate inter-modulation frequencies.
  - Undesirable harmonics generated outside the mobile radio band cause interference to adjacent services.
  - Undesirable harmonics present inside the band cause interference to other users in the mobile system.

FDMA



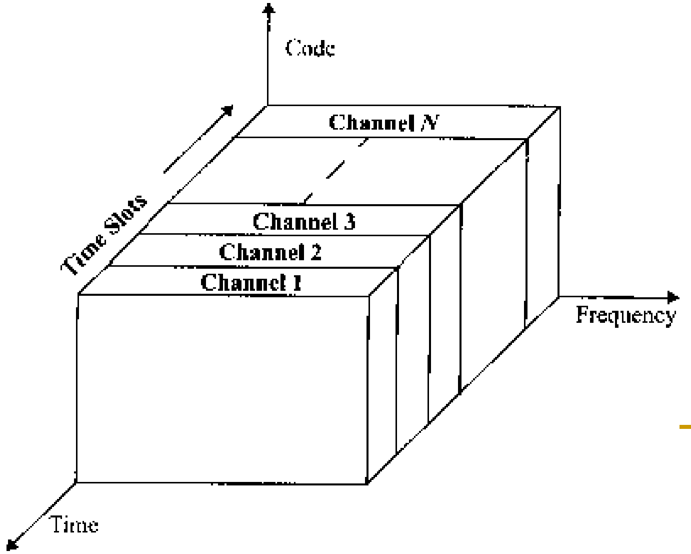
FDMA / FDD



FDMA / TDD

TDMA

- TDMA systems divide the channel time into frames. Each frame is further partitioned into time slots. In each slot only one user is allowed to either transmit or receive.
- Unlike FDMA, only digital data and digital modulation must be used.
- Each user occupies a cyclically repeating time slot, so a channel may be thought of as a particular time slot of every frame, where *N* time slots comprise a frame.



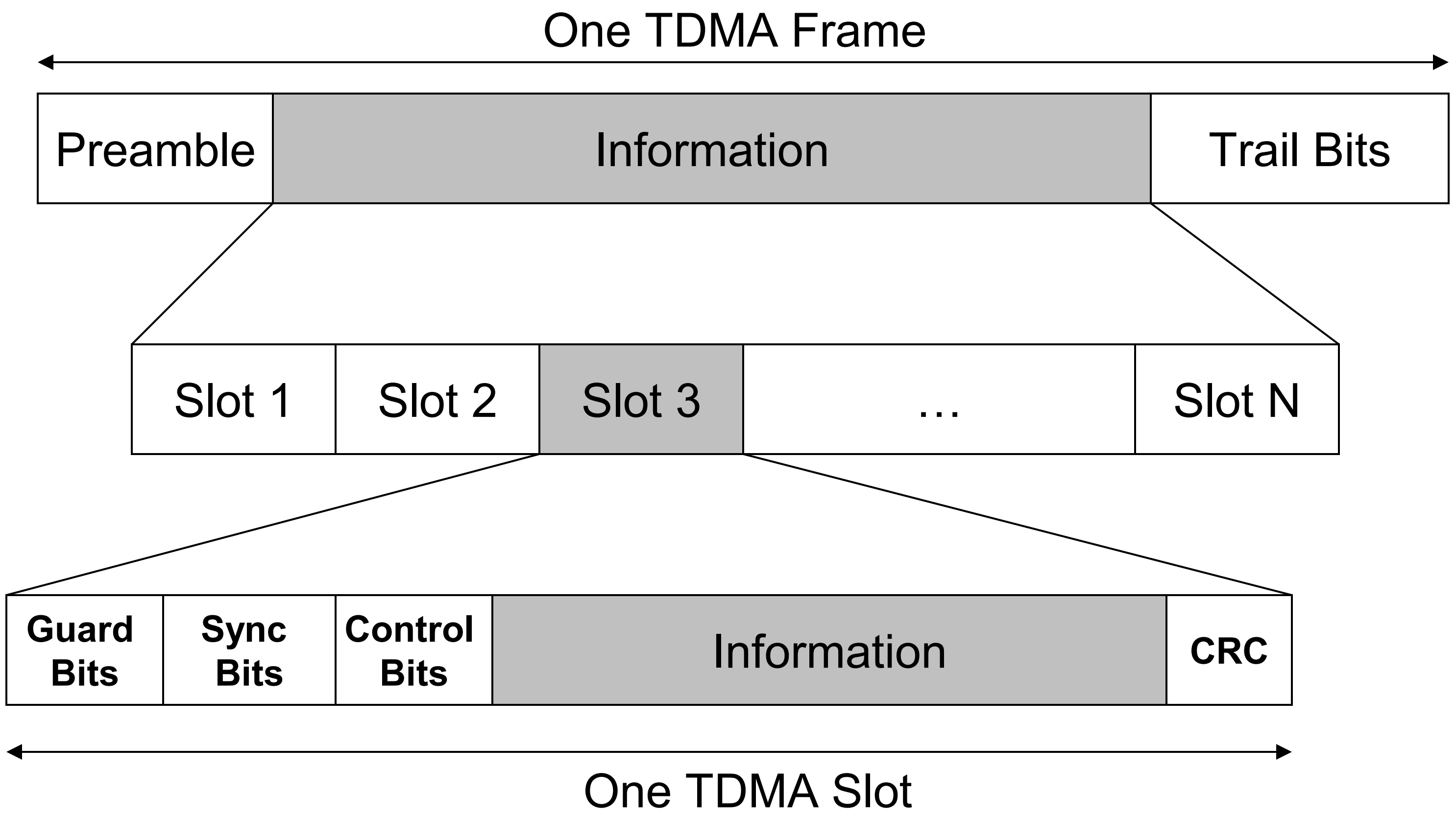
Features

- Multiple channels per carrier or RF channels.
- Burst transmission since channels are used on a timesharing basis. Transmitter can be turned off during idle periods.
- Narrow or wide bandwidth – depends on factors such as modulation scheme, number of voice channels per carrier channel.
- High ISI – Higher transmission symbol rate, hence resulting in high ISI. Adaptive equalizer required.

Features

- High framing overhead – A reasonable amount of the total transmitted bits must be dedicated to synchronization purposes, channel identification. Also guard slots are necessary to separate users.

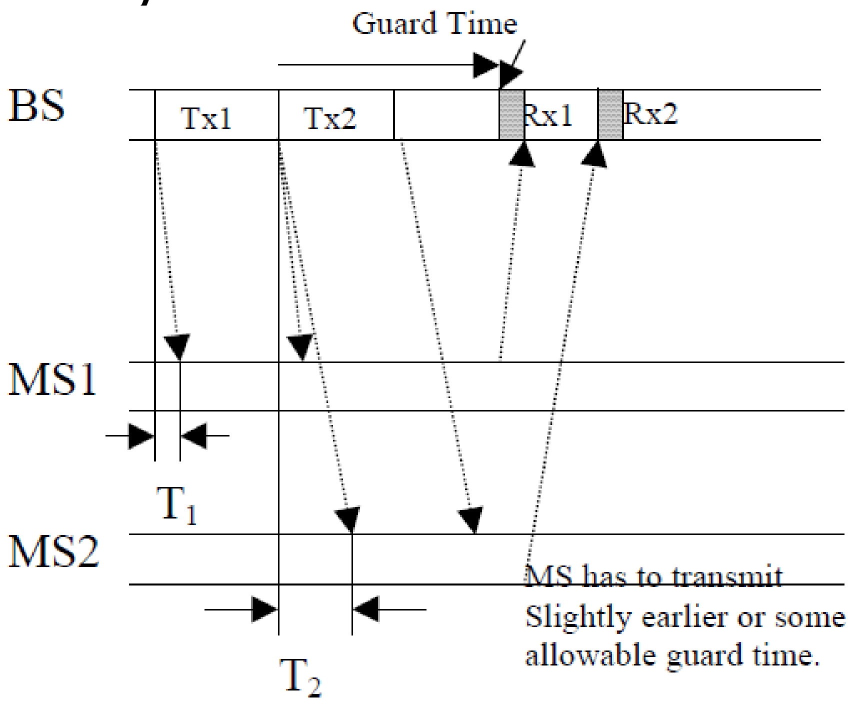
TDMA Frame



A Frame repeats in time

Features

- A guard time between the two time slots must be allowed in order to avoid interference, especially in the uplink direction. All mobiles should synchronize with BS to minimize interference.



Features

- The use of digital technology permits the inclusion of several facilities in the mobile unit, increasing its complexity. One example is the use of slow frequency hopping to counteract multi-path fading.
- Flexible data rates by assigning multiple time slots to different users based on their demand.
- No duplexer used since uses different time slots for transmission and reception. Even if FDD is used, a switch rather than a duplexer is required.
- Lower BS cost : the use of multiple channels per carrier channel provides a proportional reduction of the equipment at the BS.

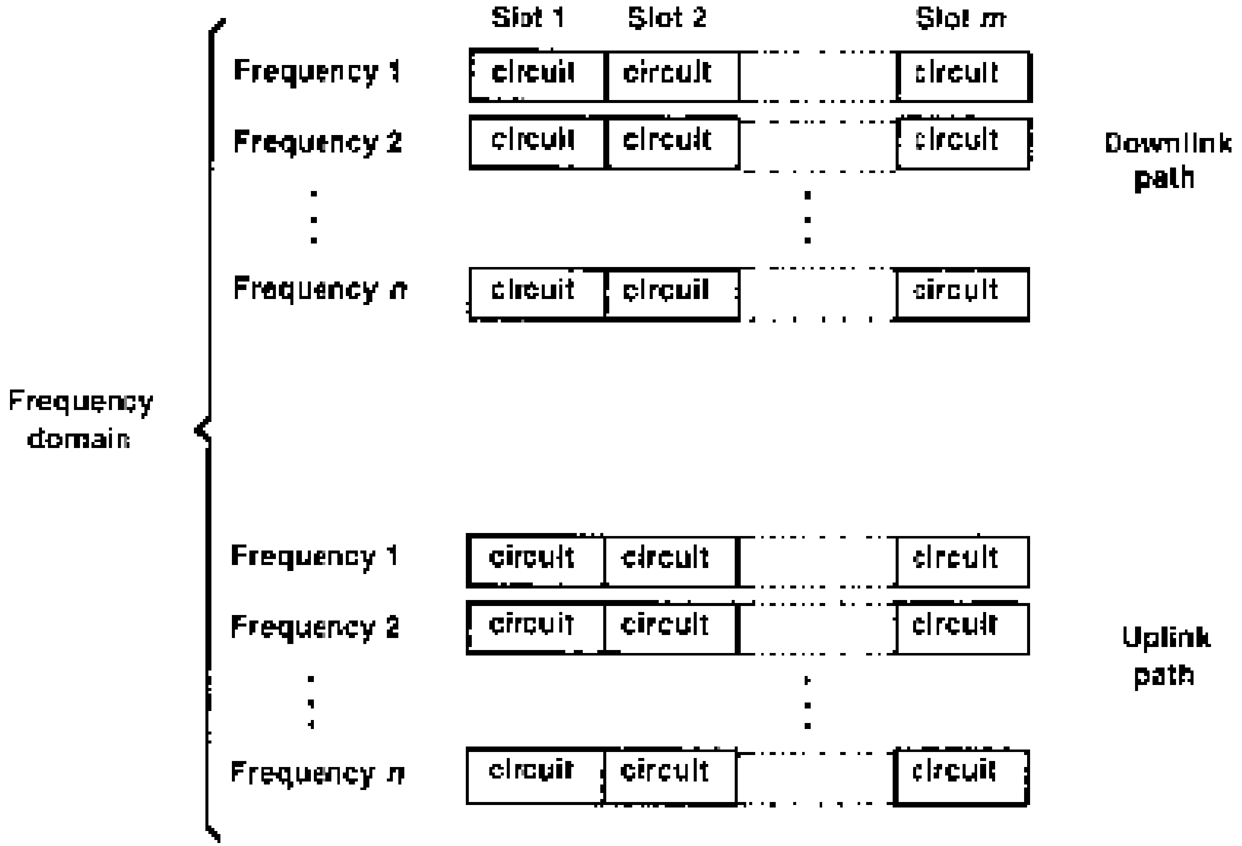
Features

- Efficient power utilization : FDMA systems require a 3- to 6-dB power back off in order to compensate for inter-modulation effects.
- Efficient handoff : TDMA systems can take advantage of the fact that the transmitter is switched off during idle time slots to improve the handoff procedure. An enhanced link control, such as that provided by mobile assisted handoff (MAHO) can be carried out by a subscriber by listening to neighboring base station during the idle slot of the TDMA frame.

TDMA/TDD and TDMA/FDD

- In TDMA/TDD system, half of the time slots in the frame information message would be used for the forward link channels and half would be used for reverse link channels. Same channel conditions.
- In TDMA/FDD systems, same frame structure can be used for both forward and reverse transmission but carrier frequencies used are different.

TDMA/TDD and TDMA/FDD



FDMA/TDMA/FDD

Synchronous and asynchronous TDMA

- Previously we mentioned the same slot of all the frames are assigned to the same user. The slot assignment can actually be fixed or dynamic.
- If the assigned slot is fixed from frame to frame for the duration of the connection, the users have to synchronize to their respective assigned slots. This mode of TDMA is referred to as STDMA.
- With packet-switched transmission, it is more efficient to allow a user to transmit only when it has packets to send. Transmission slots are dynamically assigned from frame to frame. This mode is known as ATDMA.
- A TDMA is implemented using a reservation access mechanism.

SSMA

- Spread spectrum systems : The desired signal is transmitted over a bandwidth which is much larger than the Nyquist bandwidth. It is first developed for military applications for
  - Security
  - Undetectability: minimum probability of being detected
  - Robust against intentional jammers

Applications

- Security
- Robust against unintentional interference
- It is not bandwidth efficient when used by a single user but has the capability to overcome narrowband jamming signals (cannot overcome AWGN or wideband jamming signal) and multi-path.
- Providing multiple access
- If many users can share the same spread spectrum bandwidth without interfering with one another, bandwidth efficient improved but will affect the capability to overcome jamming.

Spread Spectrum Access

- Two techniques
  - Frequency Hopped Multiple Access (FHMA)
  - Direct Sequence Multiple Access (DSMA)
    - Also called Code Division Multiple Access – CDMA

Frequency Hopping (FHMA)

- Digital muliple access technique
- A wideband radio channel is used.
  - Same wideband spectrum is used
- The carrier frequency of users are varied in a pseudo-random fashion.
  - Each user is using a narrowband channel (spectrum) at a specific instance of time.
  - The random change in frequency make the change of using the same narrowband channel very low.

Frequency Hopping (FHMA)

- The sender receiver change frequency (calling hopping) using the same pseudo-random sequence, hence they are synchronized.
- Rate of hopping versus Symbol rate
  - If hopping rate is greather: Called *Fast Frequency Hopping*
    - One bit transmitted in multiple hops.
  - If symbol rate is greater: Called *Slow Frequency Hopping*
    - Multiple bits are transmitted in a hopping period
    - GSM and Bluetooth are example systems

Code Division Multiple Access (CDMA)

- In CDMA, the narrowband message signal is *multiplied* by a very large bandwidth signal called spreading signal (code) before modulation and transmission over the air. This is called spreading.

- CDMA is also called DSSS (Direct Sequence Spread Spectrum). DSSS is a more general term.

- Message consists of symbols
  - Has symbol period and hence, symbol rate

Code Division Multiple Access (CDMA)

- Spreading signal (code) consists of chips
  - Has Chip period and and hence, chip rate
  - Spreading signal use a pseudo-noise (PN) sequence (a pseudo-random sequence)
  - PN sequence is called a codeword
  - Each user has its own cordword
  - Codewords are orthogonal. (low autocorrelation)
  - Chip rate is oder of magnitude larger than the symbol rate.
- The receiver correlator distinguishes the senders signal by examining the wideband signal with the same time-synchronized spreading code
- The sent signal is recovered by despreading process at the receiver.

CDMA Advantages

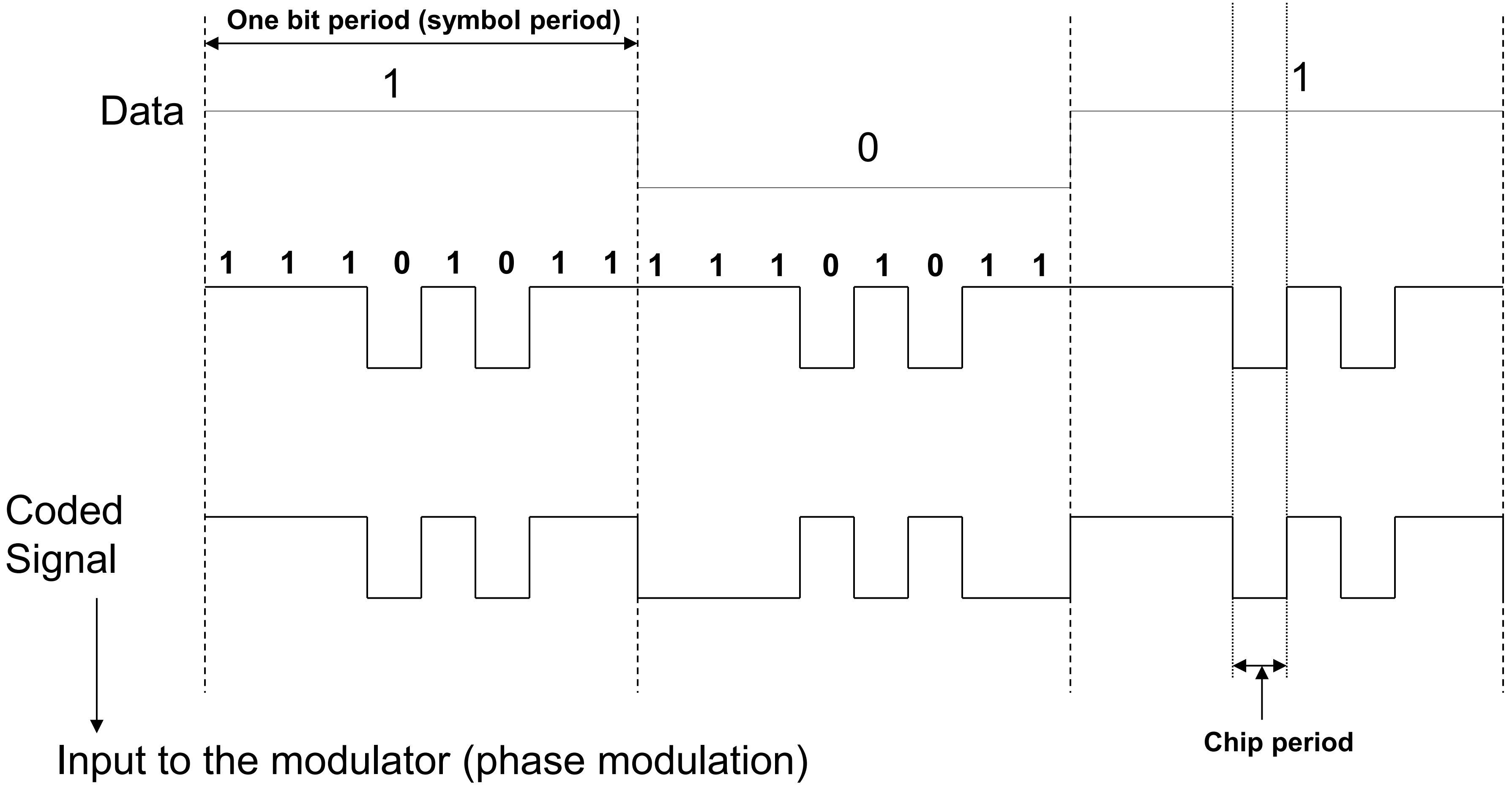
- Low power spectral density.
  - Signal is spread over a larger frequency band
  - Other systems suffer less from the transmitter
- Interference limited operation
  - All frequency spectrum is used
- Privacy
  - The codeword is known only between the sender and receiver. Hence other users can not decode the messages that are in transit
- Reduction of multipath affects by using a larger spectrum

CDMA Advantages

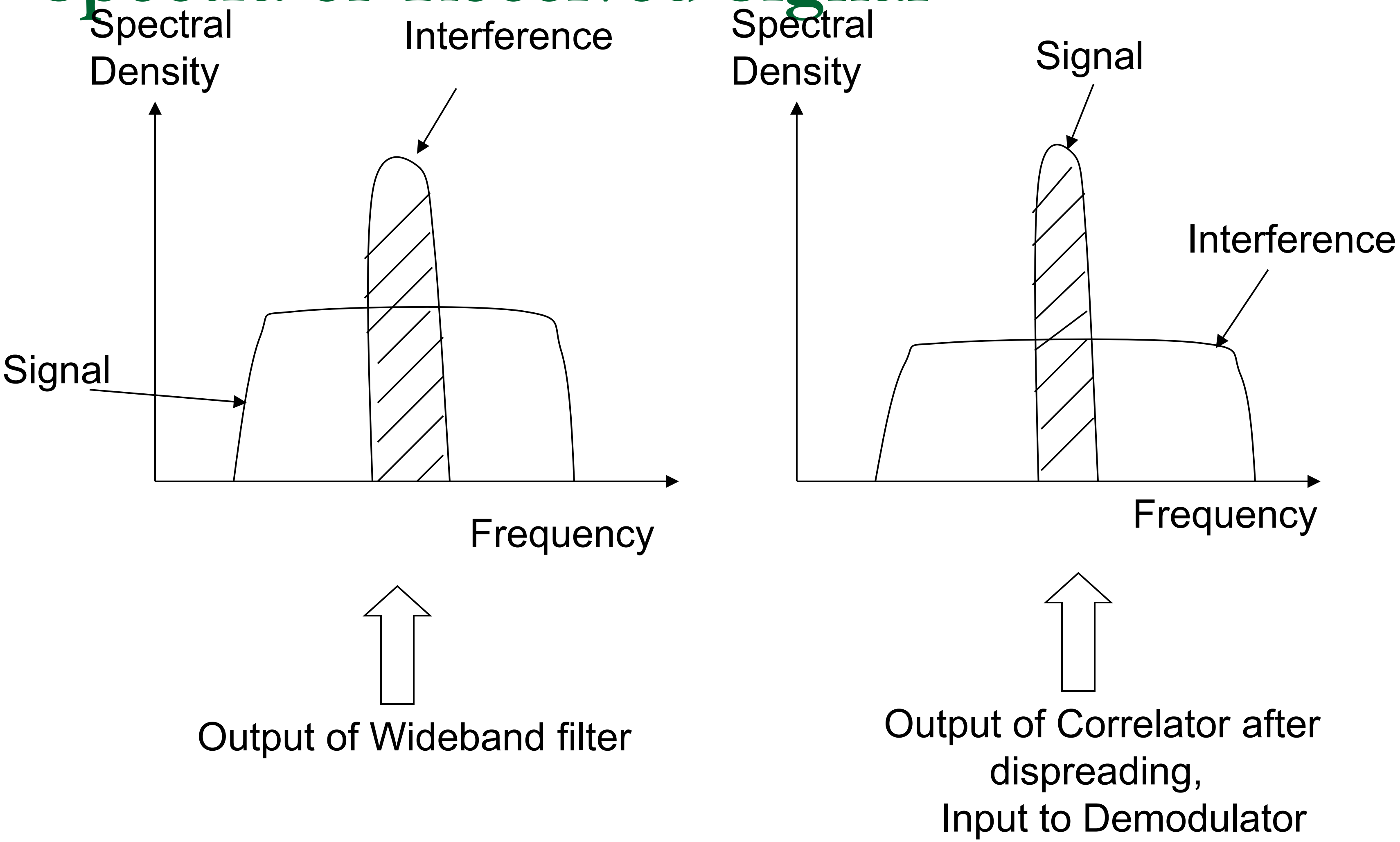
- Random access possible
  - Users can start their transmission at any time
- Cell capacity is not concerete fixed like in TDMA or FDMA systems. Has soft capacity
- Higher capacity than TDMA and FDMA
- No frequency management
- No equalizers needed
- No guard time needed
- Enables soft handoff

CDMA Principle

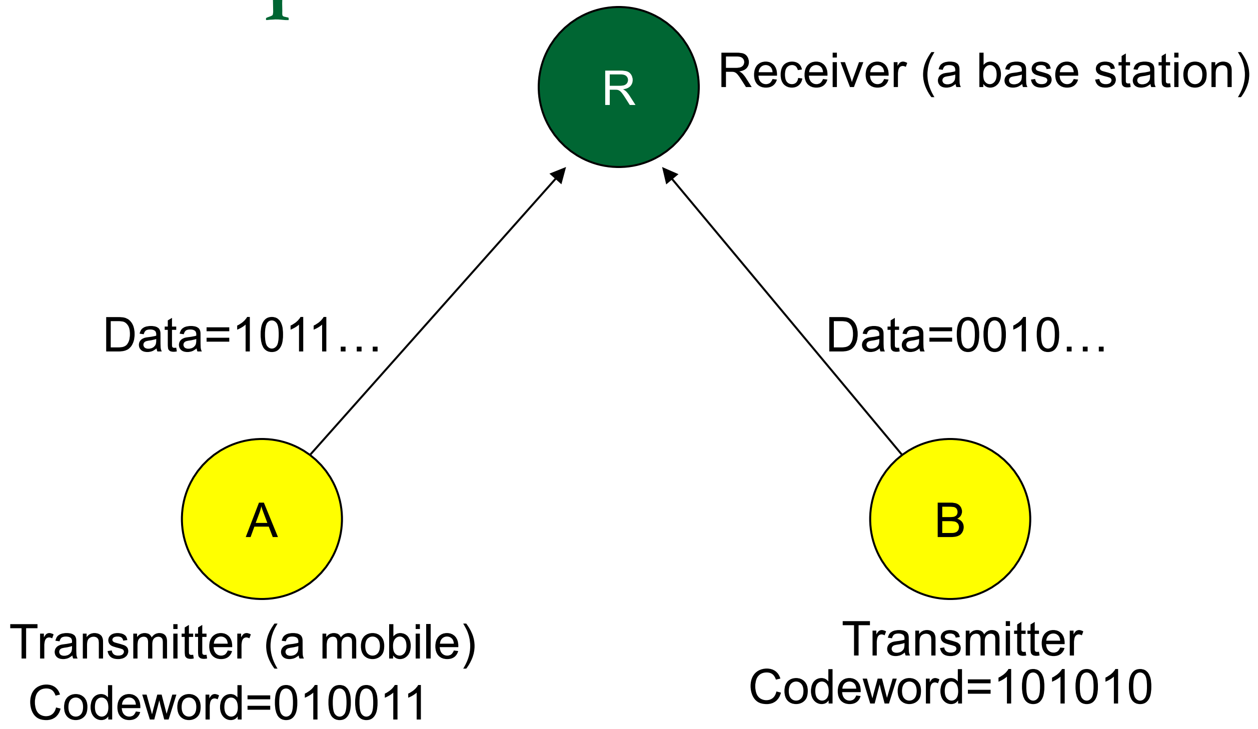
Represent bit 1 with +1
Represent bit 0 with -1



Spectra of Received Signal

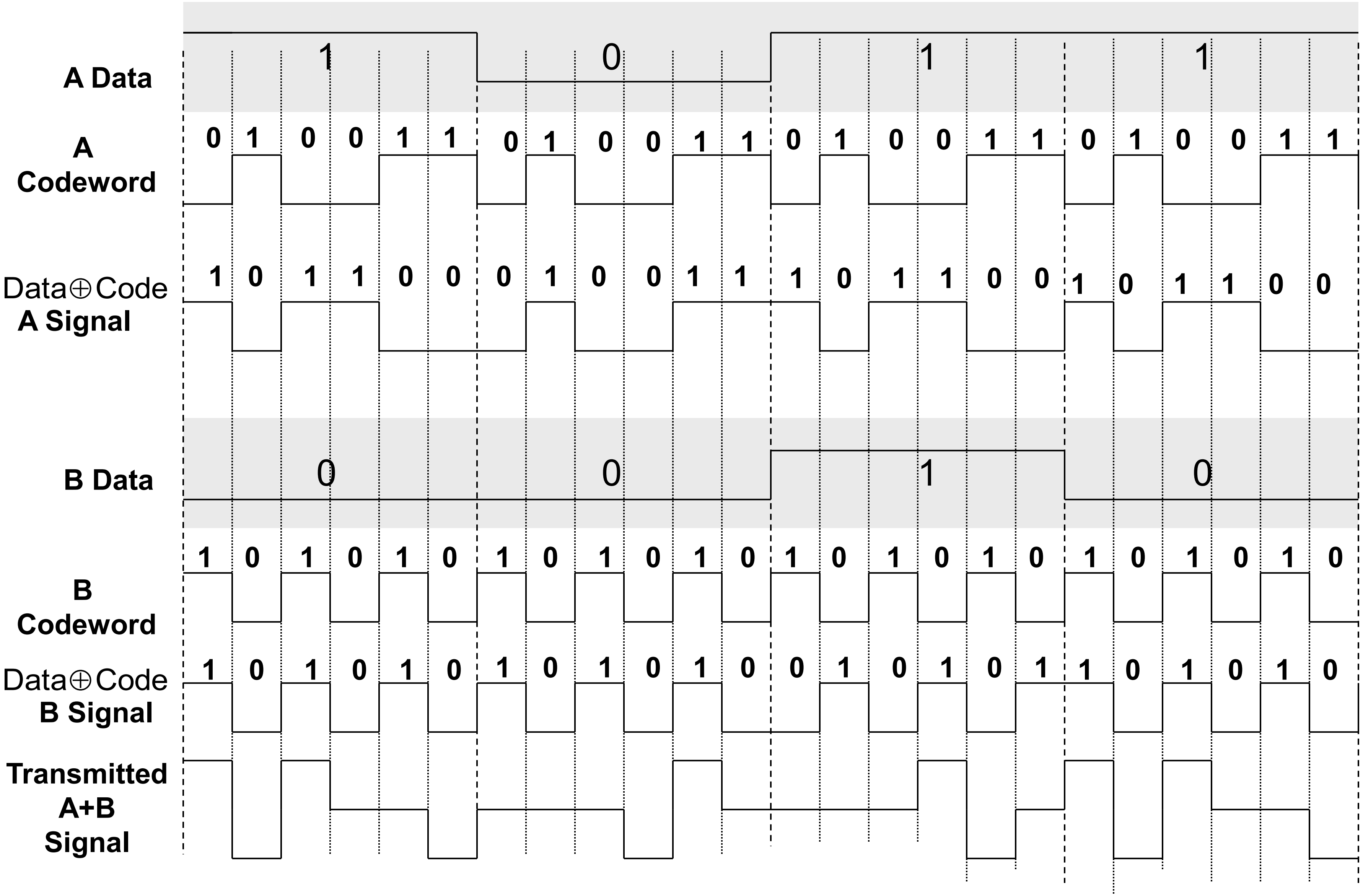


CDMA Example



Data transmitted from A and B is multiplexed using CDMA and codeword.
The Receiver de-multiplexes the data using dispreading.

CDMA Example – transmission from two sources



Hybrid Spread Spectrum Techniques

- FDMA/CDMA
  - Available wideband spectrum is frequency divided into number narrowband radio channels. CDMA is employed inside each channel.
- DS/FHMA
  - The signals are spread using spreading codes (direct sequence signals are obtained), but these signal are not transmitted over a constant carrier frequency; they are transmitted over a frequency hopping carrier frequency.

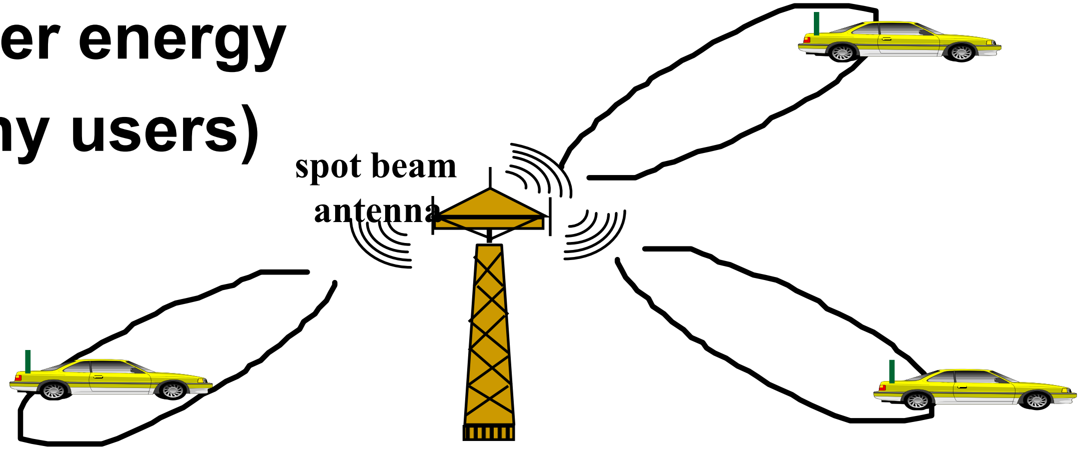
Hybrid Spread Spectrum Techniques

- Time Division CDMA (TCDMA)
  - Each cell is using a different spreading code (CDMA employed between cells) that is conveyed to the mobiles in its range.
  - Inside each cell (inside a CDMA channel), TDMA is employed to multiplex multiple users.
- Time Division Frequency Hopping
  - At each time slot, the user is hopped to a new frequency according to a pseudo-random hopping sequence.
  - Employed in severe co-interference and multi-path environments.
    - Bluetooth and GSM are using this technique.

SDMA

- Use spot beam antennas
- The different beam area can use TDMA, FDMA, CDMA
- Sectorized antenna can be thought of as a SDMA
- Adaptive antennas can be used in the future (simultaneously steer energy

in the direction of many users)



Features:

- A large number of independently steered high-gain beams can be formed without any resulting degradation in SNR ratio.
- Beams can be assigned to individual users, thereby assuring that all links operate with maximum gain.
- Adaptive beam forming can be easily implemented to improve the system capacity by suppressing co channel interference.